Department of Statistics 2023-24



Two-step Maximum Likelihood and Bayesian Estimation of Structural Equation Modelling

Candidate Number: 31518

Submitted for the Master of Science

London School of Economics

University of London

Table of Contents

Summary **v**

1 Introduction **1**
- 1.1 Background . . . 1
- 1.2 Research Motivation . . . 2
- 1.3 Outline . . . 4

2 Literature Review **5**
- 2.1 Maximum Likelihood Approach . . . 5
- 2.2 Bayesian Approach . . . 6

3 Models and Variables **8**
- 3.1 Measurement Model . . . 8
- 3.2 Structural Model . . . 9

4 Methodology **11**
- 4.1 Joint Estimation . . . 11
  - 4.1.1 Joint Maximum Likelihood Estimation . . . 11
  - 4.1.2 Joint Bayesian Estimation . . . 12
- 4.2 Proposed Two-step Estimation . . . 14
  - 4.2.1 Estimation for the First Step . . . 14
  - 4.2.2 Estimation for the Second Step . . . 16

5 Simulation Study **19**
- 5.1 Estimated Mean Analysis . . . 20
- 5.2 Estimated Standard Error Analysis . . . 23
- 5.3 Distribution of Measurement Parameters Analysis . . . 28

6 Empirical Study **31**
- 6.1 One Country Research: Germany . . . 32
- 6.2 Cross-national Research . . . 33

7 Conclusion and Discussion **38**
- 7.1 Key Findings . . . 38
- 7.2 Further Research Directions . . . 38

Bibliography **40**

Appendix **44**

List of Figures

1 The path diagram representing the example of learning habits and academic performance. 2
2 The path diagram of the simulation model. 19
3 Density plots of posterior draws of l, σ_2^2 for ML and Bayesian estimation with different R_m^2, controlling for $R_s^2 = 0, n = 200$. 29
4 Density plots of posterior draws of l, σ_2^2 for ML and Bayesian estimation with different n, controlling for $R_m^2 = 0.5, R_s^2 = 0.8$. 29
5 Density plots of posterior draws of l, σ_2^2 for ML and Bayesian estimation with different R_s^2, controlling for $R_m^2 = 0.5, n = 200$. 30
6 Estimated mean values and 95% confidence intervals of regression coefficients of 31 countries. 36
7 The heat map of average wlconDSCs level of 31 countries. 37
8 Estimated mean values and 95% confidence intervals of the covariance between benefDSC and costsDSC of 31 countries. 45

List of Tables

1 Survey questions on students' studying habits. 1
2 The conjugate prior and posterior distribution of parameters $\mathbf{\Psi_{1k}}$, Λ_k^{T}, $\mathbf{\Psi_{2k}}$, Γ_k^{T}, $\mathbf{\Phi}$. . . 13
3 True values of free loadings l, measurement error variance σ_1^2, σ_2^2, structural coefficients $\beta_1, \beta_2, \beta_3$, structural error variance ψ_1^2, ψ_2^2 of models with different R^2. 20
4 Bias of the structural parameter β_1 of two one-step methods and three two-step methods in 27 settings. 21
5 Root Mean Square Error (RMSE) of the structural parameter β_2 of two one-step methods and three two-step methods in 27 settings. 23
6 Simulation Standard Deviation vs Mean of Estimated Standard Error of β_1 in 27 settings for Joint ML and Joint Bayesian estimation methods. 24
7 Simulation Standard Deviation vs Mean of Estimated Standard Error of β_1 in 27 settings for ML + ML, ML + draws + ML, Bayesian + ML estimation methods. . . . 25
8 Coverage of 95% confidence interval of $\beta_1, \beta_2, \beta_3$ for Bayes + ML estimation approach. 27
9 Simulation results for standard error estimates of estimated structural regression coefficients β_1, β_2, β_3: average % of the variance of SAM estimator accounted for by step-2 variance alone. 27
10 The interpretation of indicators. 31
11 Estimated loading values of 4 models via SAM and joint ML methods. 33
12 Estimates for structural parameters of 31 Countries. 35
13 Root Mean Square Error (RMSE) of the structural parameter β_3 of two one-step methods and three two-step methods in 27 settings. 44

Summary

Latent variable models have been used in a wide range of fields in sociology. These models are particularly useful when the variables of interest cannot be directly observed but can be inferred from related indicators. By utilizing latent variables, researchers can discover underlying structures, patterns, and relationships that may not be immediately apparent through direct measurement. In the estimation of latent variable models, the one-step method is generally used, where both the measurement and structural models are estimated simultaneously. This study considers a two-step approach, known as structural-after-measurement (SAM) method. The first step of SAM method involves exclusively estimating the measurement model, and the second step estimates the full model with the measurement model parameters held fixed. In this research, structural equation models (SEM), where both latent variables and indicators are continuous, are used as a case study. The results demonstrate that SAM offers both practical and conceptual advantages over the one-step estimation, as well as provides comparable accuracy to the one-step method in most cases. Furthermore, this study introduces alternative two-step procedures that can serve as practical substitutes for SAM under certain conditions, offering more convenient implementation while maintaining accuracy.

Keywords: latent variable model; structural equation model; structural-after-measurement; interpretational confounding.

1 Introduction

1.1 Background

In the field of social sciences, many phenomena of interest, such as abilities, emotions and attitudes, are often not directly observable. To address this challenge, latent variables have been introduced in statistics, allowing for the quantification of abstract constructs. The idea of latent variables dates back to early 20th century statistical and psychological research. A seminal application of this concept was proposed by Spearman (1904), who observed that the results of various intelligence-related tests, such as memory, weight discrimination, and mathematical reasoning, typically exhibited significant positive correlations. These correlations were attributed to a common underlying general intelligence factor, known as the g factor. Although the g factor is not directly observable, Spearman confirmed its existence as a measure of an individual's intellectual ability.

As the term suggests, the value of a latent variable cannot be directly observed and therefore requires observable indicators for measurement. The models established for this purpose are known as measurement models. For example, as abstract constructs, students' learning habits can be measured by aggregating scores from five related questions (in Table 1), while the academic performance can be measured by grades of each subject and course participation rate. In this example, the latent variables are learning habits and academic performance, while the indicators are scores from five questions in Table 1, subject grades and course participation rate. The methods for measuring latent variables had been widely debated until 1947, when Thurstone introduced multiple factor analysis. Thurstone proposed that the relationship between measurement items (such as questionnaire items) and factors is linear, with the coefficients representing these linear relationships being referred to as factor loadings. This linear measurement relationship has since become the dominant framework in modern factor analysis.

Table 1: Survey questions on students' studying habits.

Question Number	Question Content
1	I have a fixed study time every day.
2	I like to use charts and notes while studying.
3	I often review what I have learned after studying.
4	I prefer to study in a quiet environment.
5	I often set study goals and try to achieve them.

Note: Each question is rated on a scale of 1 to 5, where 1 = "Strongly Disagree" and 5 = "Strongly Agree".

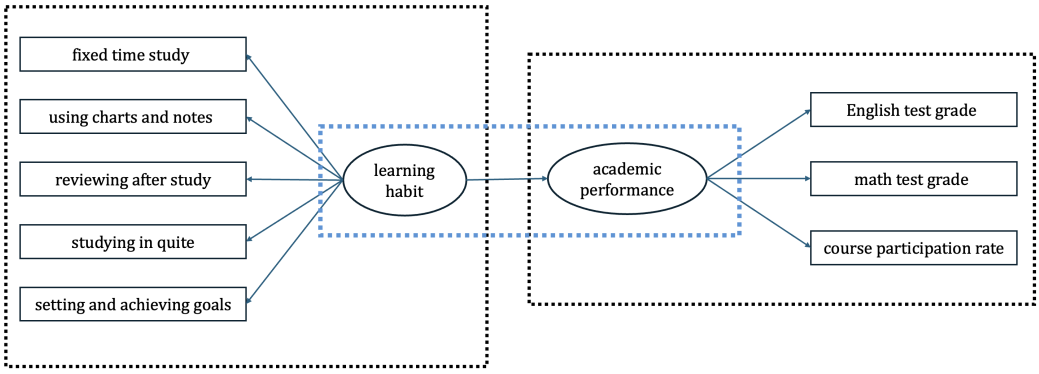
The relationship among latent variables themselves, or between latent and other observable variables, also constitutes a primary focus of interest, referred to as structural models. An early form of structural modelling is path analysis (Wright, 1934), which uses path diagrams to depict and estimate causal relationships among variables, providing a detailed discussion on the computation and interpretation of path coefficients. This regression-based approach has evolved into contemporary structural models.

Traditional path analysis typically involves only observable variables, without incorporating latent

variables. Jöreskog (1970), who systematically established structural equation modeling (SEM), integrated measurement models and path analysis to incorporate latent variable modeling. Continuing the previous example, learning habits should have a significant impact on academic performance. Thus, in a structural model, an arrow could be drawn from learning habits to academic performance to illustrate this relationship.

The advent of SEM established the general framework for latent variable models, which consist of both measurement models and structural models. Figure 1 illustrates the complete SEM framework for the previously mentioned example, where the portion enclosed in black dashed lines represents the measurement model, and the section enclosed in blue dashed lines represents the structural model. SEM assumes that variables are continuous and normally distributed. As the field has evolved, latent variable models have diversified based on the data types of indicators and latent variables. Accordingly, the relationship between variables has expanded from linear to generalised linear.

Figure 1: The path diagram representing the example of learning habits and academic performance.



Note: The variables within rectangular boxes are observable, and the variables within circular boxes are latent.

1.2 Research Motivation

The most commonly used method for estimating latent variables is estimating both the measurement model and structural model simultaneously, known as joint estimation or one-step estimation. However, in recent years, researchers have attempted to explore step-wise methods that estimate the two sub-models in multiple steps. Compared with one-step analysis, step-wise methods provide some key advantages. Firstly, high computational complexity can be mitigated, as each step contains fewer parameters compared to the joint estimation model. Additionally, Burt (1973) observed that if different models use the same measurement model and estimate all parameters simultaneously, the factor loadings of the latent variable may vary depending on the model. These variations can alter the meaning or interpretation of the latent variable. Step-wise approaches can avoid this confusion. For instance, in the example illustrated in Figure 1, the measurement relationship between learning habits and five related questions should not be influenced by academic performance. However, if the measurement and structural parts are fitted concurrently, the causal relationship between learning habits and academic performance could affect the measurement relationship. Moreover, if there is

misspecification in the structural model's assumptions, it may substantially distort the measurement parameters. Consequently, the concept and specification of the model can become ambiguous and confusing under joint estimation.

To address these deficiencies, the three-step approach was initially proposed. In the first step, only the parameters of the measurement model are estimated. The second step is to predict the latent variable values for all samples based on the estimated parameters of the measurement model. In the third step, the latent variables are treated as known and are used to fit the structural model. In this process, the measurement error is not accounted for and as a result, the estimates of the structural model parameters are generally biased. Bolck et al. (2004), Wang et al. (2019) and other researchers employed various techniques to manage and correct the measurement error arising from the use of predicted values. Nevertheless, the estimation and adjustment process is typically quite complex and cumbersome.

In this study, an alternative estimation method distinct from the one-step and three-step approaches for latent variable models is proposed. This two-step estimation method integrates the advantages of both the one-step and three-step approaches. In its first step, it mirrors the first step of the three-step estimation by only fitting the measurement model, ignoring the structural part. The second step parallels the one-step approach by estimating both the measurement and structural model simultaneously. However, in this step, the parameters of the measurement model are fixed at their estimated values from the first step. Compared with joint estimation, this method avoids the issue of interpretational confounding and is significantly less computationally demanding for both steps. Furthermore, it allows for more efficient use of data. For example, a sample with missing values in the structural variables can still contribute to the estimation of the measurement model. In contrast, under under the joint approach, such a sample can only be discarded from estimation. Compared with the three-step method, the two-step method simplifies the model framework by eliminating one step, while maintaining result accuracy and computational efficiency.

Latent variable models have derived different forms based on the data types of indicators and latent variables. Theoretically, all these forms can be applied within the two-step estimation framework. This research primarily examines the two-step analysis within the context of structural equation models, where both indicators and latent variables are assumed to be continuous. Five variations of the two-step method are developed based on the two-step estimation procedure, which attempt either classical estimation (maximum likelihood, ML) or Bayesian estimation in each step. A simulation study is conducted to compare the fit of the one-step method with several two-step approaches and examine the distribution of parameters under ML and Bayesian estimation. Additionally, a real data experiment investigates the two key advantages of the two-step analysis method over the one-step method: mitigating interpretational confounding and reducing computational complexity.

1.3 Outline

The remainder of this dissertation is organised as follows: Section 2 reviews the literature on the two-step methods under both ML and Bayesian statistical approaches. Section 3 illustrates the structural equation model specification considered in the study. Section 4 describes the definition and properties of both joint estimation and proposed two-step estimation methods. This section also discusses the implementation of these approaches using ML and Bayesian estimation. Section 5 details the simulation study, while Section 6 presents the empirical experiment. Both studies were implemented using the open-source software R package lavaan (Rosseel, 2012) and blavaan (Merkle and Rosseel, 2018). Finally, Section 7 summarises the dissertation. Bibliography and supplementary appendices are listed separately at the end.

2 Literature Review

The dominant approach to SEMs has been the one-step estimation method, also regarded as joint estimation, which simultaneously fits both the measurement and structural models. This approach is prevalent in both frequentist (Bollen, 1989) and Bayesian frameworks (Lee, 2007). While the one-step method is generally considered straightforward, Burt (1973, 1976) argued that it can lead to interpretational confounding. This occurs when the implied definition of a latent variable is partly influenced by variables that should be conceptually distinct from it. The confounding would be exacerbated with relatively weak measurement relationships or strong structural relationships. Levy (2017) noted that the joint approach can result in interpretational confounding due to covariates or outcomes in the structural model. Changes to any submodel necessitate a refitting of the full model. This approach presents not only conceptual shortcomings but also practical challenges, as it makes the estimation process computationally demanding.

In response to these issues, step-wise approaches have been proposed and attempted in recent years. The following sections will elaborate on these approaches from the perspectives of both ML and Bayesian estimation.

2.1 Maximum Likelihood Approach

Among all frequentist-based estimation methods, the most widely used in one-step SEM is the full information maximum likelihood (FIML) (Koopmans et al., 1950) estimation. Bollen (1989) provided an example to illustrate the maximum likelihood estimation (MLE) of a single parameter model of SEM. Ferron and Hess (2007) illustrated a more complex and concrete example of using MLE to solve SEM. This method is also commonly applied to other types of latent variable models, such as latent class models (Dayton and Macready, 1988), where both the indicators and latent variables are categorical, and latent trait models (Embretson (Whitely), 1984), where indicators are categorical and latent variables are continuous. The specific estimation process is presented in Section 4.1.1.

Although Burt (1976) suggested using the step-wise approach to avoid interpretational confounding, it has been more commonly applied to model selection instead of estimation. Some researchers (Anderson and Gerbing, 1988; Bollen, 2000) proposed a two-step method for model-selection purpose. The first step is fitting the measurement model, and only if the parameter significance is acceptable, the second step which is fitting the full model will be executed. In this study, the model is assumed to be correctly specified, and the proposed two-step method is primarily for estimation.

In the step-wise methods used for estimation, the naive three-step method is widely employed. This method involves initially fitting the measurement model, then predicting and assigning values to the latent variables for each unit, and finally fitting the structural model with these latent variables treated as known. For certain mdoel settings, this method does not introduce bias, as shown by Skrondal and Laake (2001), who applied it to factor analysis measurement models, a simplified SEM. Lu and

Thomas (2008) extended this idea to latent class analysis. For more general models, the latent variable values assigned in the second step do not accurately represent the true values defined in the first step, leading to a measurement error problem. The bias can be avoided using some bias-corrected three-step methods (Bolck et al., 2004; Bakk et al., 2013, Devlieger et al., 2016). Another method frequently used in SEM, but employing a different frequentist approach, is the two-stage least squares (2SLS) estimation, which is based on principles of instrumental variable estimation. Various version of the method have been discussed by Jöreskog and Sörbom (1986), Lance et al. (1988) and Bollen (1996). 2SLS was designed to address specific types of endogeneity problems, and it offers limited help in reducing interpretational confounding.

The two-step method advocated in this paper involves first fitting only the measurement model, then fitting the full model by fixing parameters of measurement model at their estimated values from the first step. Burt (1976) mentioned this idea, but a systematic approach to SEM has only been developed in recent years. Skrondal and Kuha (2012) introduced a two-step pseudo-ML method for generalized linear models to correct covariate measurement error in regression models through two-step estimation. It is quite similar to the advocated two-step method in SEM, differing only slightly in the division of parameters between the first and second steps. In the field of latent variable models, Bandeen-Roche et al. (1997) firstly proposed the application of two-step method for latent class models. Later, Xue and Bandeen-Roche (2002) refined this approach by verifying the consistency of the estimation results and establishing the asymptotic distribution of the parameters. Bakk and Kuha (2018) expanded on these earlier proposals, detailing two-step modeling as a general approach for latent class analysis, and extended it to latent trait models in 2024. Rosseel and Loh (2022) proposed a general two-step method for application in SEM: structural-after-measurement (SAM), in which the global SAM is essentially a two-step ML estimation. This approach is also compatible with other classical statistical estimation method, such as weighted least squares (WLS), generalized least squares (GLS), and unweighted least squares (ULS). This topic will be discussed in detail in Section 4.2.

The two-step method for latent variable models can now be implemented using common software, such as Latent Gold for latent class anlaysis (Vermunt and Magidson, 2021) and R package lavaan for SEM (Rosseel, 2012). Implementing the two-step method for latent trait models requires a combination of R and Mplus (Kuha and Bakk, 2024).

2.2 Bayesian Approach

In social sciences, latent variables such as attitudes and values are usually measured using ordered categories or dichotomous scales (Kaplan, 2009). Consequently, the data in such cases do not actually meet the assumption of normal distribution, and using MLE can lead to biased estimates of the mean and standard deviation (Olsson et al., 2000). As a result, the application of Bayesian methods in social sciences has been increasingly considered. Additionally, Bayesian methods perform better in estimating latent variables in certain complex models, such as those with random effect factor loadings and random slopes (Muthén, 2010), or in models containing both continuous and categorical latent variables (Depaoli, 2021). Furthermore, Bayesian statistics does not rely on asymptotic theory,

which is particularly helpful in small sample situations (Kaplan and Depaoli, 2013). It also involves generating posterior draws of parameters, which is more convenient when presenting certain statistics, such as in the confidence interval (CI) (Scheines et al., 1999).

Bayesian statistics incorporate background information when estimating model parameters. Their rapid development in recent years is largely due to the discovery of various Markov Chain Monte Carlo (MCMC) methods, which allow for the flexible generation of large samples from the posterior distribution. Early attempts to apply the Bayesian framework to latent variables were primarily focused on factor analysis, a simplified version of SEM (Martin and McDonald, 1975). As statistical software implementing MCMC methods has matured, the application of the Bayesian approach to various forms of latent variable models has become increasingly widespread. Arminger and Muthén (1998) proposed its application to a nonlinear latent variable model with polynomial latent form, using Gibbs sampling and the Metropolis-Hastings algorithm to estimate the posterior distribution of parameters and latent variables. Lee and Shi (2000) introduced the application of Bayesian statistics to SEM models, incorporating additional fixed covariates on both indicators and latent variables. He found that standard error estimates and goodness-of-fit measures can be obtained more easily compared with the ML method. Goldstein and Browne (2002) described the general structure for using MCMC to estimate multilevel factor analysis with binary responses.

The one-step analysis method used in this research follows the framework of Lee (2007), utilizing conjugate prior and Gibbs-sampling algorithm, with specific details presented in Section 4.1.2. Bayesian-based two-step analysis in latent variable models appears relatively rare, with only Levy (2023) systematically applying it in SEM, using Bayesian estimation at each step. He employed the posterior samples of measurement parameters in the second step, actually implementing the two-step method for multiple times. Simulation experiments were conducted in statistical software, including the R blavaan package (Merkle et al., 2021), Stan (Carpenter et al., 2017), and the R2jags package (RStudio Team, 2021). The two-step analysis in the MCMC context will be explained in Section 4.2.

3 Models and Variables

In the general form of a latent variable model with sample size n, assume that there are M latent variables (factors) and J observable indicators used to measure these latent variables. For a record of unit $i = 1, ..., n$, denote $\boldsymbol{\xi_i} = (\xi_{i1}, ..., \xi_{iM})^{\mathrm{T}}$ as the vector of latent variables and $\boldsymbol{Y_i} = (Y_{i1}, ..., Y_{iJ})^{\mathrm{T}}$ as the vector of indicators. The model sometimes considers including other observed variables, denoted as $\boldsymbol{Z_i} = (\boldsymbol{Z_{ip}}, \boldsymbol{Z_{io}})$ where $\boldsymbol{Z_{ip}}$ is the explanatory variable vector of $\boldsymbol{\xi_i}$ and $\boldsymbol{Z_{io}}$ is the response vector for $\boldsymbol{Z_{ip}}$ and $\boldsymbol{\xi_i}$. To clarify, $\boldsymbol{Z_{ip}}$ consists of exogenous variables, values of which are able to affect but unable to be affected by other variables within the model. On the other hand, $\boldsymbol{Z_{io}}$ consists of endogenous variables, whose values are influenced by other variables. Based on the definition, the full model form can be considered as equation (1):

$$p(\boldsymbol{Y_i}, \boldsymbol{Z_{io}}, \boldsymbol{\xi_i} | \boldsymbol{Z_{ip}}; \boldsymbol{\theta}) = p(\boldsymbol{Y_i} | \boldsymbol{Z_i}, \boldsymbol{\xi_i}; \boldsymbol{\theta_1}) \times p(\boldsymbol{Z_{io}}, \boldsymbol{\xi_i} | \boldsymbol{Z_{ip}}; \boldsymbol{\theta_2}), \tag{1}$$

where $p(\cdot|\cdot)$ denotes a conditional distribution and $\boldsymbol{\theta} = (\boldsymbol{\theta_1}, \boldsymbol{\theta_2})$ represents the vector of all parameters of interest, partitioned into two parts: $\boldsymbol{\theta_1}$, the parameter vector in the measurement model, and $\boldsymbol{\theta_2}$, the parameter vector in the structural model. On the right side of equation (1), the first probability refers to the form of measurement model part and the second probability refers to the form of structural model part. To avoid confusion between causal and correlational relationships among variables, i.e., interpretational confounding, assume that the indicators $\boldsymbol{Y_i}$ measure factors $\boldsymbol{\xi_i}$ without the affection by observed variables in the structural model. Thus, $p(\boldsymbol{Y_i} | \boldsymbol{Z_i}, \boldsymbol{\xi_i}; \boldsymbol{\theta_1}) = p(\boldsymbol{Y_i} | \boldsymbol{\xi_i}; \boldsymbol{\theta_1})$. Then the full model framework is updated as

$$p(\boldsymbol{Y_i}, \boldsymbol{Z_{io}}, \boldsymbol{\xi_i} | \boldsymbol{Z_{ip}}; \boldsymbol{\theta}) = p(\boldsymbol{Y_i} | \boldsymbol{\xi_i}; \boldsymbol{\theta_1}) \times p(\boldsymbol{Z_{io}}, \boldsymbol{\xi_i} | \boldsymbol{Z_{ip}}; \boldsymbol{\theta_2}), \tag{2}$$

implying a model for observed variables by integrating the latent factors

$$p(\boldsymbol{Y_i}, \boldsymbol{Z_{io}} | \boldsymbol{Z_{ip}}; \boldsymbol{\theta}) = \int p(\boldsymbol{Y_i} | \boldsymbol{\xi_i}; \boldsymbol{\theta_1}) \times p(\boldsymbol{Z_{io}}, \boldsymbol{\xi_i} | \boldsymbol{Z_{ip}}; \boldsymbol{\theta_2}) \, d\boldsymbol{\xi_i}. \tag{3}$$

In SEM, it is commonly regulated that the indicator $\boldsymbol{Y_i}$ and latent factors $\boldsymbol{\xi_i}$ are continuous. Additionally, in the measurement model, $\boldsymbol{\xi_i}$ are assumed to be normally distributed and each indicator in $\boldsymbol{Y_i}$ is permitted to measure only certain confirmatory latent factors in $\boldsymbol{\xi_i}$. The structural model is recursive, and the form of relationships between $\boldsymbol{\xi_i}$ and $\boldsymbol{Z_i}$ is not restricted, although this study specifically explores linear regression in structural model.

3.1 Measurement Model

The measurement model describes the path of J indicators measuring M latent variables. For the sake of conciseness, each element in $\boldsymbol{Y_i}$ is standardised to have mean 0. This standardisation omits the need for the intercept terms in the linear form of the measurement model expression. Thus the framework

of the measurement part in formula (2) is defined as

$$\boldsymbol{Y_i} = \Lambda \boldsymbol{\xi_i} + \boldsymbol{\epsilon_i}, \tag{4}$$

where Λ is a $J \times M$ matrix of loadings and $\boldsymbol{\epsilon_i}$ is the measurement error vector of length J which is independent of $\boldsymbol{\xi_i}$. Suppose $\boldsymbol{\epsilon_i}$ is distributed as $N(0, \boldsymbol{\Psi_1})$, where $\boldsymbol{\Psi_1}$ is a $J \times J$ diagonal matrix of error variance. The indicators $\boldsymbol{Y_i}$ given $\boldsymbol{\xi_i}$ are independent of each other, which is typically referred to as local independence assumption. The parameters in the measurement part are thereby $\Lambda, \boldsymbol{\Psi_1}$, collected in $\boldsymbol{\theta_1}$. From (4), the model-implied mean vector $\boldsymbol{\mu} = \mathrm{E}(\boldsymbol{Y})$ and variance-covariance matrix $\boldsymbol{\Sigma} = \mathrm{Var}(\boldsymbol{Y})$ are derived:

$$\boldsymbol{\mu} = \Lambda \mathrm{E}(\boldsymbol{\xi}), \tag{5}$$

$$\boldsymbol{\Sigma} = \Lambda \mathrm{Var}(\boldsymbol{\xi}) \Lambda^{\mathrm{T}} + \boldsymbol{\Psi_1}. \tag{6}$$

As mentioned earlier, since one latent variable is measured by a subset of indicators and each indicator is restricted to measuring certain latent factors, several zero entries exist in Λ. This measurement model setup is regarded as the general framework of confirmatory factor analysis (CFA).

3.2 Structural Model

The structural model is established to analyse the complex relationships among latent and other observed variables. SEM requires the indicators and latent factors to be continuous, but it does not impose strict requirements on the data type of observed response and explanatory variables. The form of the model as well as structural parameters $\boldsymbol{\theta_2}$ can be flexibly adjusted depending on the type of these variables. For example, logistic regression can be used with binary response $\boldsymbol{Z_{io}}$, while normal or log-normal linear regression can be applied to continuous $\boldsymbol{Z_{io}}$. To simplify the study, focus on the scenario where the model takes the form of linear regression without interactions. Update the notation for the observed variables involved in the structural model, i.e., $\boldsymbol{Z_i}$, to latent variables that are measured only by their own values without measurement error. Consequently, the symbols $\boldsymbol{Z_i}$ can be discarded, and all variables within this model can be represented using the notation $\boldsymbol{\xi_i}$. Then partition $\boldsymbol{\xi_i}$ into two sub-vectors $\boldsymbol{\eta_i}$ and $\boldsymbol{\omega_i}$ corresponding to endogenous and exogenous variables. Assume that the lengths of $\boldsymbol{\eta_i}$ and $\boldsymbol{\omega_i}$ are M_1 and M_2 respectively with the equation $M_1 + M_2 = M$ satisfied. The structural model can thus be explicitly written as

$$\boldsymbol{\eta_i} = \mathrm{B}_1 \boldsymbol{\eta_i} + \mathrm{B}_2 \boldsymbol{\omega_i} + \boldsymbol{\zeta_i}, \tag{7}$$

where B_1 is an $M_1 \times M_1$ matrix of structural parameters that govern the relationship among the endogenous latent variables, and B_2 is an $M_1 \times M_2$ regression matrix that relates the endogenous and exogenous latent variables. It is assumed that $\boldsymbol{\omega_i}$ follows $N(0, \boldsymbol{\Phi})$, where $\boldsymbol{\Phi}$ is positive definite. The structural disturbances $\boldsymbol{\zeta_i}$ are assumed to follow $N(0, \boldsymbol{\Psi_2})$, and they are independent of $\boldsymbol{\omega_i}$ and $\boldsymbol{\eta_i}$. To avoid a direct edge from a latent variable to itself, the diagonal entries of B_1 are constrained to be zero. Additionally, $\mathrm{I}_{M_1} - \mathrm{B}_1$ should be invertible to ensure the validity of expressing $\boldsymbol{\eta_i}$ as a function

of $\boldsymbol{\omega_i}$ and $\boldsymbol{\zeta_i}$ by rewriting equation (7) as $\boldsymbol{\eta_i} = (\mathrm{I}_{M_1} - \mathrm{B}_1)^{-1}(\mathrm{B}_2\boldsymbol{\omega_i} + \boldsymbol{\zeta_i})$.

The structural model specifies the joint distribution of latent variables $\boldsymbol{\eta_i}$ and $\boldsymbol{\omega_i}$, which are jointly normal with $\mathrm{E}(\boldsymbol{\xi}) = \mathrm{E}((\boldsymbol{\omega}^{\mathrm{T}}, \boldsymbol{\eta}^{\mathrm{T}})^{\mathrm{T}}) = 0$ and

$$\mathrm{Var}(\boldsymbol{\xi}) = \mathrm{Cov}((\boldsymbol{\omega}^{\mathrm{T}}, \boldsymbol{\eta}^{\mathrm{T}})^{\mathrm{T}}) = \begin{bmatrix} \mathrm{E}(\boldsymbol{\omega}\boldsymbol{\omega}^{\mathrm{T}}) & \mathrm{E}(\boldsymbol{\omega}\boldsymbol{\eta}^{\mathrm{T}}) \\ \mathrm{E}(\boldsymbol{\eta}\boldsymbol{\omega}^{\mathrm{T}}) & \mathrm{E}(\boldsymbol{\eta}\boldsymbol{\eta}^{\mathrm{T}}) \end{bmatrix}, \tag{8}$$

where

$$\begin{aligned}
\mathrm{E}(\boldsymbol{\omega}\boldsymbol{\omega}^{\mathrm{T}}) &= \boldsymbol{\Phi}, \\
\mathrm{E}(\boldsymbol{\omega}\boldsymbol{\eta}^{\mathrm{T}}) &= \boldsymbol{\Phi}\mathrm{B}_2{}^{\mathrm{T}}((\mathrm{I}_{M_1} - \mathrm{B}_1)^{-1})^{\mathrm{T}}, \\
\mathrm{E}(\boldsymbol{\eta}\boldsymbol{\omega}^{\mathrm{T}}) &= (\mathrm{I}_{M_1} - \mathrm{B}_1)^{-1}\mathrm{B}_2\boldsymbol{\Phi}, \\
\mathrm{E}(\boldsymbol{\eta}\boldsymbol{\eta}^{\mathrm{T}}) &= (\mathrm{I}_{M_1} - \mathrm{B}_1)^{-1}(\mathrm{B}_2\boldsymbol{\Phi}\mathrm{B}_2{}^{\mathrm{T}} + \boldsymbol{\Psi_2})((\mathrm{I}_{M_1} - \mathrm{B}_1)^{-1})^{\mathrm{T}}.
\end{aligned}$$

Thereby the parameters in the structural part are $(\mathrm{B}_1, \mathrm{B}_2, \boldsymbol{\Psi_2}, \boldsymbol{\Phi})$, collected in $\boldsymbol{\theta_2}$.

To identify the parameters, some parameter constraints are needed to fix the scale of $\boldsymbol{\xi_i}$. There are generally two methods to achieve identification. The first method is selecting one indicator for each latent variable and fixing its loading on the latent variable to 1. This approach anchors the interpretation in the items $\boldsymbol{Y_i}$, leaving the distribution of $\boldsymbol{\xi_i}$ free. The second method sets the variance of each latent variable to 1. However, this approach provides a slightly less straightforward interpretation of latent scales, as in this way the variance of $\boldsymbol{\xi_{im}}$ does not appear in the structural model (see covariance matrix (8)), which is the focus of the second step of estimation. Therefore, in the subsequent simulation and empirical study sections, the first method will be used to fix the scale of latent factors.

4 Methodology

This section introduces estimation methods for latent variable models. Section 4.1 presents the one-step methods under ML and Bayesian estimation, the most commonly used approaches to fit SEMs. Section 4.2 discusses the idea of the advocated two-step method. It also explores different versions of this kind of two-step approach by combining ML and Bayesian estimation in different steps.

4.1 Joint Estimation

Estimation of parameters can be undertaken in various methods. The dominant approach to conducting SEM is the joint or one-step estimation method, in a way that parameters in the measurement and structural parts of the model are estimated simultaneously. Most SEM-related publications still use classic statistical methods, i.e., frequentist estimation. However, there has been a steady increase in use of Bayesian estimation during recent years. This section introduces the joint estimation approaches to SEM using the most typical frequentist estimation method, ML, as well as Bayesian estimation.

4.1.1 Joint Maximum Likelihood Estimation

Assume that $(\boldsymbol{Y_i}, \boldsymbol{Z_i})$ are observed variables for unit $i = 1, ..., n$, being independent of each other. The log-likelihood function of model (3) is

$$l(\boldsymbol{\theta}) = \sum_{i=1}^{n} \ln p(\boldsymbol{Y_i}, \boldsymbol{Z_{io}} | \boldsymbol{Z_{ip}}; \boldsymbol{\theta}) = \sum_{i=1}^{n} \ln \int p(\boldsymbol{Y_i} | \boldsymbol{\xi_i}; \boldsymbol{\theta_1}) p(\boldsymbol{Z_{io}}, \boldsymbol{\xi_i} | \boldsymbol{Z_{ip}}; \boldsymbol{\theta_2}) \, d\boldsymbol{\xi_i}. \tag{9}$$

If any variables in the data set are missing, there are assumed to be missing at random and their contributions are omitted from log-likelihood function (9). In the case specified by the measurement model (4) and the structural model (7), given a set of i.i.d. data vectors $y = (y_1, y_2, ..., y_n)$, MLE then seeks values of $\boldsymbol{\theta}$ that maximise the multivariate normal log-likelihood given by:

$$l(\boldsymbol{\theta}|y) = -\frac{nJ}{2}\ln(2\pi) - \frac{n}{2}\ln|\boldsymbol{\Sigma}| - \frac{n}{2}\mathrm{tr}\left(S\boldsymbol{\Sigma}^{-1}\right) - \frac{n}{2}\left(\bar{y} - \boldsymbol{\mu}\right)^{\mathrm{T}} \boldsymbol{\Sigma}^{-1} \left(\bar{y} - \boldsymbol{\mu}\right), \tag{10}$$

where $\bar{y}$ is the sample mean vector, S is the sample variance-covariance matrix, $|\cdot|$ refers to the determinant of a matrix, and $\mathrm{tr}(\cdot)$ is the trace operator that computes the sum of the diagonal elements of its matrix argument. $\boldsymbol{\Sigma}$ and $\boldsymbol{\mu}$ are given in equation (5) and (6). Maximising (10) is equivalent to minimising the normal theory-based ML discrepancy function (Browne and Arminger, 1995):

$$\mathrm{F_{NML}}(\boldsymbol{\theta}) = \left(\bar{y} - \boldsymbol{\mu}\right)^{\mathrm{T}} \boldsymbol{\Sigma}^{-1} \left(\bar{y} - \boldsymbol{\mu}\right) + \mathrm{tr}\left(S\boldsymbol{\Sigma}^{-1}\right) - \ln|S\boldsymbol{\Sigma}^{-1}| - J. \tag{11}$$

Modern statistical software typically employs numerical methods (e.g., Newton-Raphson, Fletcher-Powell, Gauss-Newton, Levenberg-Marquardt) to minimise equation (11). The algorithm begins with random initial values for each parameter and undergoes multiple iterations and adjustments until

equation (11) reaches its minimum value. This study uses R package lavaan, which defaults to the Quasi-Newton method for numerical optimization. It is noted that MLE is a frequentist statistical approach, and therefore its optimal properties requires a sufficiently large sample.

Significance tests are conducted in common ways, e.g., Wald tests (z-tests) for individual coefficients and likelihood ratio tests for one or more coefficients.

4.1.2 Joint Bayesian Estimation

Bayesian theorem provides a systematic method to update the probabilities with prior information and observed data. This is summarised as the posterior probability $P(\boldsymbol{\theta}|\boldsymbol{Y})$ proportional to the product of likelihood function $P(\boldsymbol{Y}|\boldsymbol{\theta})$ and prior probability $P(\boldsymbol{\theta})$, i.e., $P(\boldsymbol{\theta}|\boldsymbol{Y}) \propto P(\boldsymbol{Y}|\boldsymbol{\theta})P(\boldsymbol{\theta})$. The following section introduces the distribution forms of the priors $P(\boldsymbol{\theta})$ commonly used in Bayesian SEM and the algorithm for generating samples of posteriors distribution $P(\boldsymbol{\theta}|\boldsymbol{Y})$.

For clarity and consistency, a simple matrix transformation is applied to the structural model function (7):

$$\boldsymbol{\eta_i} = (\mathrm{I}_{M_1} - \mathrm{B}_1)^{-1}\mathrm{B}_2\boldsymbol{\omega_i} + (\mathrm{I}_{M_1} - \mathrm{B}_1)^{-1}\boldsymbol{\zeta_i}. \tag{12}$$

Denote that $\Gamma = (\mathrm{I}_{M_1} - \mathrm{B}_1)^{-1}\mathrm{B}_2$, $\boldsymbol{\delta_i} = (\mathrm{I}_{M_1} - \mathrm{B}_1)^{-1}\boldsymbol{\zeta_i}$ and its covariance matrix as $\boldsymbol{\Psi_{bayes}} = (\mathrm{I}_{M_1} - \mathrm{B}_1)^{-1}\boldsymbol{\Psi_2}((\mathrm{I}_{M_1} - \mathrm{B}_1)^{-1})^{\mathrm{T}}$. Hence, the two submodels (4) and (12) are summarised as the multivariate normal distribution $\boldsymbol{Y_i} \sim N(\Lambda\boldsymbol{\xi_i}, \boldsymbol{\Psi_1})$ and $\boldsymbol{\eta_i} \sim N(\Gamma\boldsymbol{\omega_i}, \boldsymbol{\Psi_{bayes}})$.

Bayesian estimation of $\boldsymbol{\theta}$ and $\boldsymbol{\xi} = (\boldsymbol{\xi_1}, ..., \boldsymbol{\xi_n})$ is obtained with Markov Chain Monte Carlo (MCMC) methods. The idea is that each parameter starts from an initial value and undergoes a random walk process to generate a Markov chain. The stationary distribution of the chain is the posterior distribution of the parameter. Specifically, a sequence of random observations from the joint posterior distribution $(\boldsymbol{\theta}, \boldsymbol{\xi}|\boldsymbol{Y})$ is generated via the Gibbs sampler. The process begins with the initial values $(\boldsymbol{\theta}^{(0)}, \boldsymbol{\xi}^{(0)})$, and at the rth iteration, using the current values $(\boldsymbol{\theta}^{(r-1)}, \boldsymbol{\xi}^{(r-1)})$, the Gibbs sampler is carried out repeatedly with the following stages until unknown variables converge to their posterior distribution:

a. Generate random variates $\boldsymbol{\xi}^{(r)}$ from the conditional distribution $p(\boldsymbol{\xi}|\boldsymbol{\theta}^{(r-1)}, \boldsymbol{Y})$;

b. Generate random variates $\boldsymbol{\theta}^{(r)}$ from the conditional distribution $p(\boldsymbol{\theta}|\boldsymbol{\xi}^{(r)}, \boldsymbol{Y})$.

The deduction of the posteriors in **a** and **b** is demonstrated in the following.

In Gibbs sampler **a**, $\boldsymbol{\xi}$ is treated as random variables rather than parameters so that no prior is set for $\boldsymbol{\xi}$. For $i = 1, \ldots, n$, $\boldsymbol{\xi_i}$ are mutually independent, and $\boldsymbol{Y_i}$ are also mutually independent given $(\boldsymbol{\xi_i}, \boldsymbol{\theta})$, thus

$$p(\boldsymbol{\xi}|\boldsymbol{Y}, \boldsymbol{\theta}) = \prod_{i=1}^{n} p(\boldsymbol{\xi_i}|\boldsymbol{Y_i}, \boldsymbol{\theta}) \propto \prod_{i=1}^{n} p(\boldsymbol{Y_i}|\boldsymbol{\xi_i}, \boldsymbol{\theta})p(\boldsymbol{\xi_i}|\boldsymbol{\theta}).$$

Given that the conditional distribution of $\boldsymbol{\xi_i}$ given $\boldsymbol{\theta}$, and $\boldsymbol{Y_i}$ given $(\boldsymbol{\xi_i}, \boldsymbol{\theta})$ are both multivariate normal,

it can be shown that (Lindley and Smith, 1972)

$$\boldsymbol{\xi_i}|\boldsymbol{\theta}, \boldsymbol{Y_i} \sim N\left((\boldsymbol{\Sigma}_{\boldsymbol{\xi}}^{-1} + \Lambda^{\mathrm{T}}\boldsymbol{\Psi}_{\mathbf{1}}^{-1}\Lambda)^{-1}\Lambda^{\mathrm{T}}\boldsymbol{\Psi}_{\mathbf{1}}^{-1}\boldsymbol{Y_i},\ (\boldsymbol{\Sigma}_{\boldsymbol{\xi}}^{-1} + \Lambda^{\mathrm{T}}\boldsymbol{\Psi}_{\mathbf{1}}^{-1}\Lambda)^{-1}\right), \tag{13}$$

where

$$\boldsymbol{\Sigma}_{\boldsymbol{\xi}} = \begin{bmatrix} \Gamma\boldsymbol{\Phi}\Gamma^{\mathrm{T}} + \boldsymbol{\Psi}_{\boldsymbol{bayes}} & \Gamma\boldsymbol{\Phi} \\ \boldsymbol{\Phi}\Gamma^{\mathrm{T}} & \boldsymbol{\Phi} \end{bmatrix}.$$

In Gibbs sampler **b**, $\boldsymbol{\theta}$ is parameters of interests and by Bayesian theorem, $p(\boldsymbol{\theta}|\boldsymbol{\xi}, \boldsymbol{Y}) \propto p(\boldsymbol{Y}, \boldsymbol{\xi}|\boldsymbol{\theta})p(\boldsymbol{\theta})$. Assuming that the prior distribution of $\boldsymbol{\theta_1}$ and $\boldsymbol{\theta_2}$ are independent, i.e., $p(\boldsymbol{\theta}) = p(\boldsymbol{\theta_1})p(\boldsymbol{\theta_2})$. Given that $p(\boldsymbol{Y}|\boldsymbol{\xi}, \boldsymbol{\theta}) = p(\boldsymbol{Y}|\boldsymbol{\xi}, \boldsymbol{\theta_1})$ and $p(\boldsymbol{\xi}|\boldsymbol{\theta}) = p(\boldsymbol{\xi}|\boldsymbol{\theta_2})$, $p(\boldsymbol{\theta}|\boldsymbol{\xi}, \boldsymbol{Y})$ can be decomposed as follows:

$$\begin{aligned} p(\boldsymbol{\theta}|\boldsymbol{\xi}, \boldsymbol{Y}) &= p(\boldsymbol{\theta_1}, \boldsymbol{\theta_2}|\boldsymbol{\xi}, \boldsymbol{Y}) \\ &\propto \big(p(\boldsymbol{Y}|\boldsymbol{\xi}, \boldsymbol{\theta_1})p(\boldsymbol{\theta_1})\big) \times \big(p(\boldsymbol{\xi}|\boldsymbol{\theta_2})p(\boldsymbol{\theta_2})\big) \\ &= \big(p(\boldsymbol{Y}|\boldsymbol{\xi}, \Lambda, \boldsymbol{\Psi_1})p(\Lambda, \boldsymbol{\Psi_1})\big) \qquad (14) \\ &\quad \times \big(p(\boldsymbol{\eta}|\boldsymbol{\omega}, \Gamma, \boldsymbol{\Psi}_{\boldsymbol{bayes}})p(\Gamma, \boldsymbol{\Psi}_{\boldsymbol{bayes}})\big) \qquad (15) \\ &\quad \times \big(p(\boldsymbol{\omega}|\boldsymbol{\Phi})p(\boldsymbol{\Phi})\big). \qquad (16) \end{aligned}$$

The expression (14), (15), and (16) detail the Bayesian relationships between each parameter and its associated random variables, where the first $p(\cdot|\cdot)$s represent the likelihood expressions of the known variables influenced by the parameters, and the latter $p(\cdot)$s are the prior density functions of the parameters themselves.

The widely employed prior probability in the general Bayesian approach is conjugate prior distribution (Gelman et al., 1995), where the posterior distribution retains the same parametric form as the prior distribution. This property facilitates the derivation of analytical solutions, making it particularly useful for efficient parameter estimation. The likelihoods in expression (14) and (15) follow a normal distribution with unknown mean and variance. The normal distribution is a conjugate family for the mean parameter, and inverse-gamma (IG) distribution serves as a conjugate family for the variance parameter. In expression (16), the likelihood is assumed to be normal with mean of 0, and the inverse Wishart (IW) distribution is a conjugate family for the variance parameter. Details of the conjugate prior and corresponding posterior distributions are provided in Table 2 (Lee, 2007).

Table 2: The conjugate prior and posterior distribution of parameters $\boldsymbol{\Psi_{1k}}$, Λ_k^{T}, $\boldsymbol{\Psi_{2k}}$, Γ_k^{T}, $\boldsymbol{\Phi}$.

Parameter	**Prior**	**Posterior**
$\boldsymbol{\Psi_{1k}}$ $(1 \le k \le J)$	$IG(\alpha_{01k}, \beta_{01k})$	$IG(n/2 + \alpha_{01k}, \beta_{1k})$
Λ_k^{T} $(1 \le k \le J)$	$N(\Lambda_{0k}, \boldsymbol{\Psi_{1k}}\mathrm{H}_{0yk})$	$N(a_k, \boldsymbol{\Psi_{1k}}\mathrm{A}_k)$
$\boldsymbol{\Psi_{2k}}$ $(1 \le k \le M_1)$	$IG(\alpha_{02k}, \beta_{02k})$	$IG(n/2 + \alpha_{02k}, \beta_{2k})$
Γ_k^{T} $(1 \le k \le M_1)$	$N(\Gamma_{0k}, \boldsymbol{\Psi_{2k}}\mathrm{H}_{0\eta k})$	$N(g_k, \boldsymbol{\Psi_{2k}}\mathrm{G}_k)$
$\boldsymbol{\Phi}$	$IW(\mathrm{R}_0, \rho_0)$	$IW\left((\boldsymbol{\omega}\boldsymbol{\omega}^T + \mathrm{R}_0^{-1}), n + \rho_0\right)$

Note: $\boldsymbol{\Psi_{1k}}$ and $\boldsymbol{\Psi_{2k}}$ are respectively the kth diagonal elements of $\boldsymbol{\Psi_1}$ and $\boldsymbol{\Psi_{bayes}}$. Λ_k^{T}, Γ_k^{T}, $\boldsymbol{Y_k^{\mathrm{T}}}$ and $\boldsymbol{\eta_k^{\mathrm{T}}}$ are respectively the kth rows of Λ, Γ, $\boldsymbol{Y}$ and $\boldsymbol{\eta}$.
The real numbers α_{01k}, β_{01k}, ρ_0, vectors Λ_{0k}, Γ_{0k}, and positive matrices R_0, H_{0yk}, $\mathrm{H}_{0\eta k}$ are hyperparameters.
$\mathrm{A}_k = (\mathrm{H}_{0yk}^{-1} + \boldsymbol{\xi}\boldsymbol{\xi}^{\mathrm{T}})^{-1}$, $a_k = \mathrm{A}_k(\mathrm{H}_{0yk}^{-1}\Lambda_{0k} + \boldsymbol{\xi}\boldsymbol{Y_k})$, $\beta_{1k} = \beta_{01k} + 2^{-1}(\boldsymbol{Y_k^{\mathrm{T}}}\boldsymbol{Y_k} - a_k^{\mathrm{T}}\mathrm{A}_k^{-1}a_k + \Lambda_{0k}^{\mathrm{T}}\mathrm{H}_{0yk}^{-1}\Lambda_{0k})$.
$\mathrm{G}_k = (\mathrm{H}_{0\eta k}^{-1} + \boldsymbol{\omega}\boldsymbol{\omega}^{\mathrm{T}})^{-1}$, $g_k = \mathrm{G}_k(\mathrm{H}_{0\eta k}^{-1}\Gamma_{0k} + \boldsymbol{\omega}\boldsymbol{\eta_k})$, $\beta_{2k} = \beta_{02k} + 2^{-1}(\boldsymbol{\eta_k^{\mathrm{T}}}\boldsymbol{\eta_k} - g_k^{\mathrm{T}}\mathrm{G}_k^{-1}g_k + \Gamma_{0k}^{\mathrm{T}}\mathrm{H}_{0\eta k}^{-1}\Gamma_{0k})$.

In Bayesian statistics, one critical procedure is convergence test, which exams whether the MCMC simulations have converged to the target distribution. This step is vital because only when the chains have converged do the samples obtained from MCMC accurately represent the posterior distribution and are thereby valid for statistical inference and prediction. If the chains have not converged, the results might be inaccurate or biased. Common convergence tests include ploting trace plots, autocorrelation plots, checking Gelman statistic and Rhat.

4.2 Proposed Two-step Estimation

This study advocates a two-step approach, which aligns with ideas already proposed for latent variable models in some previous research (Bakk and Kuha, 2018, 2024; Rosseel and Loh, 2022; Levy, 2023): in the first stage, only the parameters of the measurement model are estimated; in the second stage, parameters of the measurement model are fixed at their estimated values from the first stage and the full model is then fitted. Rosseel and Loh (2022) employs frequency-based statistical methods, including ML, for both steps in SEM, while Levy (2023) utilizes the MCMC method for estimation in both phases. This study aims to expand on these approaches by exploring various combinations of ML and MCMC methods across different stages. In practice, prior information may exist solely for the measurement part or the structural part. Additionally, Bayesian estimation, when compared with ML in the first stage, offers richer statistical insights, which will be elaborated upon in the subsequent contents.

4.2.1 Estimation for the First Step

The first step essentially involves estimating a CFA model, which includes only the indicators and latent factors. The latent variables $\boldsymbol{\xi_i}$ now take the role of exogenous latent variables $\boldsymbol{\omega_i}$, and the distribution of $\boldsymbol{\xi_i}$ is assumed to be marginal normal, denoted as $N(0, \boldsymbol{\Phi_0})$. In this step, the parameters ought to be loadings Λ, measurement error variance $\boldsymbol{\Psi_1}$, and the latent variable variance $\boldsymbol{\Phi_0}$. However, the full model assumes that latent variables are conditionally normally distributed given exogenous variables $\boldsymbol{Z_{ip}}$, which conflicts with the assumptions of the CFA model. Therefore, the covariance matrix of latent factors $\boldsymbol{\xi_i}$ will be re-estimated in the second step. Although all Λ, $\boldsymbol{\Psi_1}$ and $\boldsymbol{\Phi_0}$ are estimated in the first step, only the estimates of Λ and $\boldsymbol{\Psi_1}$ are fixed in the second step. Note that only Λ and $\boldsymbol{\Psi_1}$ are collected in $\boldsymbol{\theta_1}$.

In the general CFA model, $\boldsymbol{\xi_i}$ is typically assumed to be multivariate, with correlations among latent factors to be estimated. In this scenario, any biases resulting from misspecifying the measurement model for one latent variable may spread to the (correctly specified) measurement models for other

latent variables within the same measurement block. To maximise the benefit of the two-step approach, it is advisable in the first step to estimate the measurement models for each latent factor separately, unless there exists concrete evidence of significant correlations among latent factors. It ensures robustness against model misspecifications in the measurement part, and increases the chances that a misspecified measurement block will not affect other measurement blocks. In simulation and empirical study sections, the strategy of estimating the measurement model for each latent factor separately will be employed in the first step.

Maximum Likelihood Estimation

If the first step estimation is conducted by ML, referring to the log-likelihood function (9) and (10) of the joint estimation and ignoring the part related to $\boldsymbol{Z_i}$, the log likelihood function of the measurement part is expressed as

$$\begin{aligned} l(\boldsymbol{\theta_1}, \boldsymbol{\Phi_0}) &= \sum_{i=1}^{n} \ln p(\boldsymbol{Y_i}|\boldsymbol{\theta_1}) = \sum_{i=1}^{n} \ln \int p(\boldsymbol{Y_i}|\boldsymbol{\xi_i}) p(\boldsymbol{\xi_i}; \boldsymbol{\Phi_0})\, d\boldsymbol{\xi_i} \\ &= -\frac{n}{2} \ln \left| 2\pi (\Lambda \boldsymbol{\Phi_0} \Lambda^{\mathrm{T}} + \boldsymbol{\Psi_1}) \right| - \frac{n}{2} \mathrm{tr} \left((\Lambda \boldsymbol{\Phi_0} \Lambda^{\mathrm{T}} + \boldsymbol{\Psi_1})^{-1} S \right), \end{aligned} \tag{17}$$

where S is the sample variance-covariance matrix. Since the measurement model for each latent factor is usually estimated separately, $\boldsymbol{\Phi_0}$ is assumed to be a diagonal matrix. The presence of integrals makes finding the maximum of the log-likelihood function (17) challenging, similar to joint estimation, and thus requires numerical methods for optimization. The point estimates $\hat{\boldsymbol{\theta}}_{\mathbf{1}}$ are the values that maximise function (17) and the estimated standard errors $sd(\hat{\boldsymbol{\theta}}_{\mathbf{1}})$ are then computed with the Hessian matrix.

Bayesian Estimation

If the first step is estimated with Bayesian statistics, similar to Section 4.1.2, the MCMC method is applied to generate posterior draws of the unknown variables $(\boldsymbol{\theta_1}, \boldsymbol{\Phi_0}, \boldsymbol{\xi_i})$ given $\boldsymbol{Y_i}$. These variables are still obtained with the Gibbs sampler via stage **a** and **b**, with $\boldsymbol{\theta}$ replaced by $\boldsymbol{\theta_1}$ and $\boldsymbol{\Phi_0}$. Similarly, since $\boldsymbol{Y_i}$ and $\boldsymbol{\xi_i}$ are both normally distributed, in Gibbs sampler **a**, the conditional posterior distribution of $\boldsymbol{\xi_i}$ given $\boldsymbol{Y_i}$, $\boldsymbol{\theta_1}$ and $\boldsymbol{\Phi_0}$ is

$$\boldsymbol{\xi_i}|\boldsymbol{\theta_1}, \boldsymbol{\Phi_0}, \boldsymbol{Y_i} \sim N\left((\Lambda^{\mathrm{T}} \boldsymbol{\Psi}_{\mathbf{1}}^{-1} \Lambda + \boldsymbol{\Phi}_{\mathbf{0}}^{-1})^{-1} \Lambda^{\mathrm{T}} \boldsymbol{\Psi}_{\mathbf{1}}^{-1} \boldsymbol{Y_i},\ (\Lambda^{\mathrm{T}} \boldsymbol{\Psi}_{\mathbf{1}}^{-1} \Lambda + \boldsymbol{\Phi}_{\mathbf{0}}^{-1})^{-1} \right).$$

Gibbs sampler **b** generates the posterior draws of Λ, $\boldsymbol{\Psi_1}$ and $\boldsymbol{\Phi_0}$. Keeping the prior of parameters the same as in the joint Bayesian approach, the posterior distribution of Λ and $\boldsymbol{\Psi_1}$ refers to the first two rows of Table 2, while the posterior distribution of $\boldsymbol{\Phi_0}$ is found in the last row, requiring only the replacement of $\boldsymbol{\omega}$ with $\boldsymbol{\xi}$.

After generating a series of posterior draws of the measurement parameter $\boldsymbol{\theta_1}$, the burn-in period (when the Markov chain has not yet reached its stationary phase) is discarded. The remaining draws follow the posterior distribution of $\boldsymbol{\theta_1}$, allowing for the computation of various statistics of the parameters.

4.2.2 Estimation for the Second Step

The procedure of second step estimation is similar to the joint approach described in Section 4.1, used to estimate the full model parameters $\boldsymbol{\theta} = (\boldsymbol{\theta_1}, \boldsymbol{\theta_2})$. After obtaining the $\hat{\boldsymbol{\theta}}_\mathbf{1}$ from the first step by either ML or Bayesian estimation, in the second step, $\boldsymbol{\theta_1}$ is treated as known parameters, whose values fixed at $\hat{\boldsymbol{\theta}}_\mathbf{1}$. Namely $\boldsymbol{\theta} = (\boldsymbol{\theta_1}, \boldsymbol{\theta_2})$ is substituted by $\boldsymbol{\theta} = (\hat{\boldsymbol{\theta}}_\mathbf{1}, \boldsymbol{\theta_2})$. There are also two optional estimation methods for the second step, ML and Bayesian approaches. Thus, several combinations of estimation methods across two steps can be attempted and discussed in the following. It is important to note that the form of parameters used in Bayesian estimation of the structural model, i.e., Γ and $\boldsymbol{\Psi_{bayes}}$, differs from that used in MLE, i.e., B_1 and B_2 and $\boldsymbol{\Psi_2}$. Therefore, simple linear transformation of the parameters is necessary when different estimation methods are employed in two steps, as referenced in the equation (12).

ML + ML (global SAM)

Start with the global SAM (Rosseel and Loh, 2022), i.e., structural after measurement method: MLE used in both steps, which is known as the pseudo-maximum likelihood (PML) estimation (Gong and Samaniego, 1981). Such two-stage estimators are consistent and exhibit an asymptotic normal distribution under broad regularity conditions, as discussed by Gourieroux and Monfort (1995, sections 24.2.4 and 24.2.2). In this study, these conditions are satisfied because firstly, both the joint estimator $\hat{\boldsymbol{\theta}}$ and step 1 estimator $\hat{\boldsymbol{\theta}}_\mathbf{1}$ of two-step method are ML estimators; secondly, $\boldsymbol{\theta_1}$ and $\boldsymbol{\theta_2}$ vary independently of each other.

The values of $\boldsymbol{\theta_2}$ that minimise the discrepancy function in (10) with $\boldsymbol{\theta_1}$ fixed at $\hat{\boldsymbol{\theta}}_\mathbf{1}$, are its point estimates $\hat{\boldsymbol{\theta}}_\mathbf{2}$. The standard errors of $\boldsymbol{\theta_2}$ can be derived by computing the Hessian matrix. However, these kind of standard errors are not accurate because the maximisation of the likelihood function only uses the estimates of the measurement parameters and does not incorporate information about their variances. As a result, the standard errors provided by the Hessian matrix underestimate $sd(\hat{\boldsymbol{\theta}}_\mathbf{2})$. In other words, the uncertainty from the first step is not conveyed to the second step, causing the variances of the parameters in the second step to exclude this portion of uncertainty. The procedure for modifying the standard error of $\boldsymbol{\theta_2}$ is illustrated as follows.

Suppose that the step 2 is based on n samples and step 1 on n_1 samples (in cases where $\boldsymbol{Y_i}$ is missing but any values of $\boldsymbol{Z_i}$ are available for the same i). Let the Fisher information matrix for $\boldsymbol{\theta}$ in the one-step model be

$$\mathbf{I}(\boldsymbol{\theta}^*) = \begin{bmatrix} \mathbf{I_{11}} & \mathbf{I_{12}} \\ \mathbf{I_{12}^T} & \mathbf{I_{22}} \end{bmatrix},$$

where $\boldsymbol{\theta}^*$ denotes the true value of $\boldsymbol{\theta}$ and the partitioning corresponds to $\boldsymbol{\theta_1}$ and $\boldsymbol{\theta_2}$. The estimated asymptotic variance matrix of one-step ML estimator $\boldsymbol{\theta}$ is thus given by $\hat{\mathbf{V}}_\mathbf{ML} = \mathbf{I}^{-1}(\hat{\boldsymbol{\theta}})/n$. Similarly, the Fisher information matrix for $\boldsymbol{\theta_1}$ can be obtained and is denoted as Σ_{11}/n_1, representing the asymptotic variance matrix of the step 1 estimator $\hat{\boldsymbol{\theta}}_\mathbf{1}$. Then the two-step corrected asymptotic

variance-covariance matrix of the structural parameters $\hat{\boldsymbol{\theta}}_{\mathbf{2}}$ is $\mathbf{V}/n$, where

$$\mathbf{V} = \mathbf{I}_{\mathbf{22}}^{-\mathbf{1}} + \mathbf{I}_{\mathbf{22}}^{-\mathbf{1}}\mathbf{I}_{\mathbf{12}}^{\mathrm{T}}\big((n/n_1)\Sigma_{11}\big)\mathbf{I}_{\mathbf{12}}\mathbf{I}_{\mathbf{22}}^{-\mathbf{1}} = \mathbf{V}_{\mathbf{2}} + \mathbf{V}_{\mathbf{1}}. \tag{18}$$

In equation (18), the factor n/n_1 adjusts for the possibly different sample sizes in the two steps (Xue and Bandeen-Roche, 2002). $\mathbf{V}_{\mathbf{2}}$ describes the variability of $\hat{\boldsymbol{\theta}}_{\mathbf{2}}$ if the step 1 parameter $\boldsymbol{\theta}_{\mathbf{1}}$ is already known, while $\mathbf{V}_{\mathbf{1}}$ accounts for the additional variability that arises from the fact that $\boldsymbol{\theta}_{\mathbf{1}}$ is actually unknown but estimated by $\hat{\boldsymbol{\theta}}_{\mathbf{1}}$. The estimated variance matrix of the two-step estimator $\hat{\boldsymbol{\theta}}_{\mathbf{2}}$ is then $\hat{\mathbf{V}}/n$ by substituting $\hat{\boldsymbol{\theta}} = (\hat{\boldsymbol{\theta}}_{\mathbf{1}}, \hat{\boldsymbol{\theta}}_{\mathbf{2}})$ for $\boldsymbol{\theta}^*$ in $\mathbf{I}_{\mathbf{22}}$, $\mathbf{I}_{\mathbf{12}}$ and Σ_{11}.

The term $\hat{\mathbf{V}}_{\mathbf{1}}/n$ is the "naive" estimator, describing the variability in $\hat{\boldsymbol{\theta}}_{\mathbf{2}}$ if $\boldsymbol{\theta}_{\mathbf{1}}$ is known, leaving out the uncertainty that comes from estimating the parameters in the measurement part. $\hat{\mathbf{V}}_{\mathbf{2}}/n$ is added to account for the two-step estimation process, without which the serious underestimation of $\mathrm{Var}(\hat{\boldsymbol{\theta}}_{\mathbf{2}})$ and consequently $sd(\hat{\boldsymbol{\theta}}_{\mathbf{2}})$ may occur. In existing software, R package lavaan (Rosseel, 2012) with "sam" command can directly output the estimated standard errors of the two-step estimator; the "naive" estimator can be obtained by applying the "sem" command twice.

ML + draws + ML, Bayesian + ML

In SAM, the procedure for estimating standard error of structural parameters is complicated and intricate. Thus some modification to SAM approach is proposed to address this issue. The idea is that the uncertainty in the first step implies that the true values of $\boldsymbol{\theta}_{\mathbf{1}}$ are not always equal to $\hat{\boldsymbol{\theta}}_{\mathbf{1}}$. By using a potential true value each time to estimate the second step and repeating this process for multiple times, the integrated results will effectively capture the uncertainty from the first step.

For the scenario where the first step uses ML, the parameters are assumed to follow a multivariate normal distribution, allowing multiple samples drawn from $N(\hat{\boldsymbol{\theta}}_{\mathbf{1}}, sd^2(\hat{\boldsymbol{\theta}}_{\mathbf{1}}))$ to represent possible values of $\boldsymbol{\theta}_{\mathbf{1}}$. When Bayesian method is employed in the first step, the loading matrix Λ is still normally distributed whereas the measurement error variance $\boldsymbol{\Psi}_{\mathbf{1}}$ follows an inverse-gamma distribution. The operational procedure is significantly simpler, as it allows for the direct use of the parameters' posterior draws as the potential values of the parameters, which is an advantage over the ML approach.

Assume that K possible values of $\boldsymbol{\theta}_{\mathbf{1}}$ have been sampled, denoted by $\hat{\boldsymbol{\theta}}_{\mathbf{11}}, ..., \hat{\boldsymbol{\theta}}_{\mathbf{1}\boldsymbol{K}}$. Subsequently, conduct the second step with each $\hat{\boldsymbol{\theta}}_{\mathbf{1}\boldsymbol{k}}$ to obtain the point estimates $\hat{\boldsymbol{\theta}}_{\mathbf{21}}$,..., $\hat{\boldsymbol{\theta}}_{\mathbf{2}\boldsymbol{K}}$ by maximising the corresponding log-likelihood function and "naive" standard error estimates $sd(\hat{\boldsymbol{\theta}}_{\mathbf{21}})$, ..., $sd(\hat{\boldsymbol{\theta}}_{\mathbf{2}\boldsymbol{K}})$ using the Hessian matrix for the measurement parameters. With the formula

$$\mathrm{Var}(\boldsymbol{\theta}_{\mathbf{2}}) = \mathrm{E}\big(\mathrm{Var}(\boldsymbol{\theta}_{\mathbf{2}}|\boldsymbol{\theta}_{\mathbf{1}})\big) + \mathrm{Var}\big(\mathrm{E}(\boldsymbol{\theta}_{\mathbf{2}}|\boldsymbol{\theta}_{\mathbf{1}})\big)$$

and the fact that the estimation of $\boldsymbol{\theta}_{\mathbf{2}\boldsymbol{k}}$ is actually conditional on $\hat{\boldsymbol{\theta}}_{\mathbf{1}\boldsymbol{k}}$ in the kth step 2, the accurate estimated variance of $\boldsymbol{\theta}_{\mathbf{2}}$ is computed as

$$\mathrm{Var}(\hat{\boldsymbol{\theta}}_{\mathbf{2}}) = \frac{1}{K}\sum_{k=1}^{K} sd^2(\hat{\boldsymbol{\theta}}_{\mathbf{2}\boldsymbol{k}}) + \frac{1}{K-1}\sum_{k=1}^{K}(\hat{\boldsymbol{\theta}}_{\mathbf{2}\boldsymbol{k}} - \bar{\hat{\boldsymbol{\theta}}}_{\mathbf{2}})^2,$$

where $\bar{\hat{\boldsymbol{\theta}}}_{\mathbf{2}} = \sum_{k=1}^{K} \hat{\boldsymbol{\theta}}_{\mathbf{2k}}/K$. The estimated standard error of structural parameters $sd(\hat{\boldsymbol{\theta}}_{\mathbf{2}})$ is therefore equal to $\sqrt{\text{Var}(\hat{\boldsymbol{\theta}}_{\mathbf{2}})}$. The point estimate $\hat{\boldsymbol{\theta}}_{\mathbf{2}}$ is directly obtained by substituting $\hat{\boldsymbol{\theta}}_{\mathbf{1}}$ into step 2 for once, regardless of whether the first step utilizes ML or Bayesian estimation.

ML + draws + Bayesian, Bayesian + Bayesian

This part explores the application of Bayesian estimation in step 2, which is similar to the joint Bayesian estimation in Section 4.1.2, albeit with some minor differences. The generation of posterior draws for latent variables $\boldsymbol{\xi_i}$ in Gibbs sampler **a** is consistent with the joint Bayesian method, as detailed in expression (13). In Gibbs sampler **b**, the measurement parameters Λ and $\boldsymbol{\Psi_1}$ are fixed at their estimated values from the first step, leaving only the generation of draws of Γ, $\boldsymbol{\Psi_{bayes}}$ and $\boldsymbol{\Phi}$, whose posterior distributions are illustrated in Table 2. The step 1 parameters $\boldsymbol{\theta_1}$ and step 2 parameters $\boldsymbol{\theta_2}$ are assumed to be independent in this study, so the posterior distributions of all parameters in $\boldsymbol{\theta_2}$ do not depend on $\boldsymbol{\theta_1}$. After generating a certain number of samples for $\boldsymbol{\theta_2}$, the initial segment, which does not reach a stationary distribution, is discarded. The remaining samples are then considered the posterior draws, which can be used to extract statistical information. To ensure that the uncertainty from the step 1 is propagated to step 2, multiple possible values of $\boldsymbol{\theta_1}$ are utilized in an attempt to execute the step 2.

However, the second step uses a complete dataset that includes $\boldsymbol{Z}$, and the presence of a large number of effective samples can significantly prolong the time it takes for the Markov chain to reach a stationary condition. Additionally, using multiple possible values of step 1 parameters $\boldsymbol{\theta_1}$ to estimate step 2 further extends the estimation time. It also requires substantial storage space to save the posterior draws for structural parameters each time. The two-step method aims to enhance estimation efficiency, but the drawbacks of Bayesian estimation in step 2 conflict with this goal. Therefore, it is generally recommended not to use Bayesian estimation in the second step of SEM. In this research, MCMC in step 2 will not be discussed in the simulation and empirical study.

5 Simulation Study

In this section, a simulation experiment is conducted to investigate the properties of the one-step and two-step estimation methods, as well as the characteristics of different two-step methods. The focus is primarily on three aspects: the estimation means and standard errors of the parameters in the structural model, and the distribution of parameters in the measurement model.

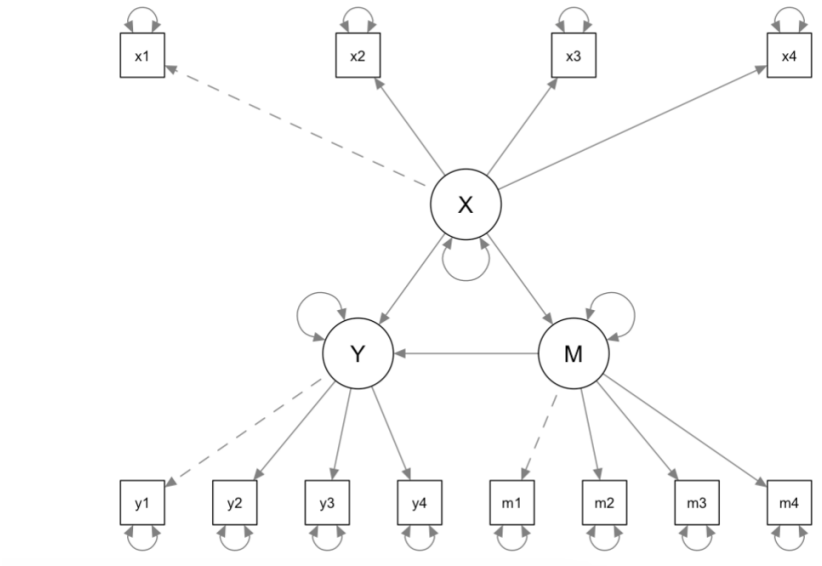
In the full model with sample size n, for each unit $i = 1, ..., n$, the measurement model contains three latent variables X_i, Y_i, M_i, each measured by four exclusive indicators. X_i is measured by $x1_i$, $x2_i$, $x3_i$ and $x4_i$; Y_i is measured by $y1_i$, $y2_i$, $y3_i$ and $y4_i$; M_i is measured by $m1_i$, $m2_i$, $m3_i$ and $m4_i$. $x1_i$, $y1_i$ and $m1_i$ are set as anchor indicators. All variables have been standardised to a mean of zero, allowing the omission of the intercept terms in the model. The measurement models for Y_i is described in equation (19), with similar models for the other two latent variable. In the structural model, there are two regression equations. One explains the relationship between two latent factors M_i and X_i (equation 21) while the other explains the relationship between the latent factor Y_i and them (equation 20). The path diagram for each simulation is shown in Figure 2.

$$(y1_i, y2_i, y3_i, y4_i)^{\mathrm{T}} = (l_{y1}, l_{y2}, l_{y3}, l_{y4})^{\mathrm{T}} \times Y_i + (\epsilon_{y1_i}, \epsilon_{y2_i}, \epsilon_{y3_i}, \epsilon_{y4_i})^{\mathrm{T}}, \tag{19}$$

$$Y_i = \beta_1 \times X_i + \beta_2 \times M_i + \delta_{Y_i}, \tag{20}$$

$$M_i = \beta_3 \times X_i + \delta_{M_i}. \tag{21}$$

Figure 2: The path diagram of the simulation model.



In this structural equation model, constrain the true values of variance of all variables equal to 1. Set true values of loadings for $y1$, $m1$ and $x1$ equal to 1, and denote their measurement error variance as σ_1^2. Suppose the true values of loading parameters for non-anchor indicators are the same, denoted by l. Thus their true measurement error variance are equal as well, denoted by σ_2^2. Set the true values of

β_1 and β_2 to be equal, and denote the regression error variance in structural model (20) and (21) as ψ_1^2 and ψ_2^2, respectively.

The simulation study framework is designed with three levels of R^2 statistics set for the measurement models, three levels for the structural models, and three different sample sizes. This setup results in a total of 27 simulation settings. Specifically, the R^2 for the measurement model is set to $R_m^2 = 0.5, 0.7, 0.9$, representing low, medium and high intensity measurement relationships. The R^2 for the structural model is set to $R_s^2 = 0, 0.4, 0.8$, indicating low, medium and high intensity structural relationships. The number of units in each dataset is set to $n = 200, n = 600, n = 1000$, suggesting small, medium and large sample size. True values of some parameters with different R^2 levels are illustrated in Table 3.

Table 3: True values of free loadings l, measurement error variance σ_1^2, σ_2^2, structural coefficients $\beta_1, \beta_2, \beta_3$, structural error variance ψ_1^2, ψ_2^2 of models with different R^2.

	measurement parameters				structural parameters		
R_m^2	l	σ_1^2	σ_2^2	R_s^2	β_1, β_2	β_3	ψ_1^2, ψ_2^2
0.5	0.707	1	0.5	0	0	0	1
0.7	0.837	0.429	0.3	0.4	0.387	0.632	0.6
0.9	0.949	0.111	0.1	0.8	0.472	0.894	0.2

For each setting, 2,000 simulated datasets are generated and estimated with two one-step approaches: joint Bayesian, joint ML, and three two-step approaches: ML + ML (global SAM), ML + draws + ML, Bayesian + ML. In approaches involving Bayesian estimation, default priors from R blavvan package are used for cases with weak prior information: loadings and regression coefficients have prior distributions $N(0, 100)$, while measurement and structural error variance have prior distributions $IG(0.001, 0.001)$. The variance of latent variables also have prior distribution of $IG(0.001, 0.001)$, as the Wishart distribution can be simplified to a gamma distribution when there is no significant correlation between variables. For each Bayesian estimation, two Markov chains are generated, each with a burn-in length of 2,000 samples and a total length of 10,000 posterior samples. Upon examination, the two chains generally converge for datasets in every setting. In the ML + draws + ML and Bayesian + ML estimation methods, after completing the first step, 2,000 samples of the measurement model parameters $\boldsymbol{\theta_1}$ are drawn, each being used in the second step.

5.1 Estimated Mean Analysis

In the study of point estimation, the ML + ML and ML + draws + ML methods yield identical point estimates. Therefore, only the joint ML, joint Bayesian, ML + ML, and Bayesian + ML estimation methods are considered for discussion. The initial focus is on the average Bias values of 2,000 estimates in each setting, as these partly reflect systematic errors. Taking the average bias of β_1 as an example (as shown in Table 4), the joint ML estimation method exhibits the smallest bias. It shows a

balanced frequency of positive and negative occurrences, with an extremely small order of magnitude in scientific notation. Combined with theoretical knowledge, this indicates that for each setting, the joint ML estimator can be considered as an unbiased estimator of the parameters. Building on this, it is evident that in almost all settings, the joint Bayesian method tends to produce higher values than the true parameter value, with larger errors. This can be attributed to the prior information causing the expectation of the parameter's posterior to be higher than the true parameter value. Additionally, the positive bias of the Bayesian + ML estimation increases as the sample size of a single dataset n increases, which indicates that the influence of prior probability becomes more pronounced with larger sample sizes. However, the influence of sample size on joint Bayesian estimation is not as significant as on the Bayesian + ML method. The ML + ML method also shows a relatively even distribution of positive and negative bias, differing from joint ML by 1 or 2 decimal places. This indicates high overall accuracy, qualifying it as an unbiased estimator. The situations for β_2 and β_3 are similar.

Table 4: Bias of the structural parameter β_1 of two one-step methods and three two-step methods in 27 settings.

		one-step		two-step	
R_m^2	R_s^2	Joint ML	Joint Bayesian	ML + ML	Bayesian + ML
$n = 200$					
0.5	0.0	9.71×10^{-18}	1.30×10^{-4}	9.71×10^{-18}	2.45×10^{-12}
	0.4	-1.00×10^{-8}	4.71×10^{-3}	3.58×10^{-8}	-1.57×10^{-1}
	0.8	-1.03×10^{-9}	3.73×10^{-3}	4.20×10^{-7}	-4.98×10^{-3}
0.7	0.0	-2.70×10^{-16}	1.01×10^{-4}	-2.70×10^{-16}	2.08×10^{-17}
	0.4	2.87×10^{-9}	1.74×10^{-3}	-2.16×10^{-8}	-3.80×10^{-2}
	0.8	3.23×10^{-9}	1.49×10^{-3}	1.02×10^{-7}	-4.33×10^{-3}
0.9	0.0	-3.01×10^{-17}	-5.84×10^{-5}	-3.01×10^{-17}	-1.06×10^{-18}
	0.4	4.19×10^{-8}	3.33×10^{-4}	-2.02×10^{-8}	-3.62×10^{-3}
	0.8	-2.58×10^{-9}	3.20×10^{-4}	-1.23×10^{-7}	-6.80×10^{-3}
$n = 600$					
0.5	0.0	-2.33×10^{-16}	-3.92×10^{-5}	-2.33×10^{-16}	-9.65×10^{-13}
	0.4	-1.18×10^{-8}	1.47×10^{-3}	3.96×10^{-8}	1.01×10^{-4}
	0.8	-1.16×10^{-9}	1.05×10^{-3}	4.08×10^{-7}	2.99×10^{-3}
0.7	0.0	-7.51×10^{-18}	-8.66×10^{-8}	-7.51×10^{-18}	-5.96×10^{-18}
	0.4	2.94×10^{-9}	5.39×10^{-4}	-1.97×10^{-8}	-1.25×10^{-3}
	0.8	2.56×10^{-9}	4.36×10^{-4}	1.03×10^{-7}	1.05×10^{-3}
0.9	0.0	6.70×10^{-18}	-1.92×10^{-5}	6.70×10^{-18}	-5.48×10^{-18}
	0.4	4.47×10^{-8}	9.86×10^{-5}	-2.22×10^{-8}	1.53×10^{-4}
	0.8	-3.16×10^{-9}	5.78×10^{-5}	-1.11×10^{-7}	2.46×10^{-4}
$n = 1000$					
0.5	0.0	-1.22×10^{-17}	-6.20×10^{-6}	-1.22×10^{-17}	-7.03×10^{-17}
	0.4	-1.07×10^{-8}	8.22×10^{-4}	3.65×10^{-8}	8.22×10^{-4}

Continued on next page

Table 4: (continued)

		one-step		two-step	
R_m^2	R_s^2	Joint ML	Joint Bayesian	ML + ML	Bayesian + ML
	0.8	-1.58×10^{-9}	5.95×10^{-4}	3.90×10^{-7}	1.74×10^{-3}
0.7	0.0	-3.09×10^{-17}	9.16×10^{-7}	-3.09×10^{-17}	9.84×10^{-17}
	0.4	3.29×10^{-9}	3.77×10^{-4}	-7.20×10^{-8}	4.37×10^{-4}
	0.8	1.79×10^{-9}	2.43×10^{-4}	9.88×10^{-8}	6.10×10^{-4}
0.9	0.0	-2.44×10^{-17}	2.92×10^{-6}	-2.44×10^{-17}	-3.34×10^{-17}
	0.4	3.91×10^{-8}	1.19×10^{-4}	-1.18×10^{-8}	9.19×10^{-5}
	0.8	-3.39×10^{-9}	7.53×10^{-5}	-1.10×10^{-7}	1.56×10^{-4}

Next, the Root Mean Square Error (RMSE) statistics is selected to analyse each parameter in each setting, which is always a positive number and provides a more intuitive measure of accuracy. Table 5 shows the RMSE of parameter β_2 under each setting. Overall, joint Bayesian is the most stable, with RMSE showing little variation across settings, benefiting from the fact that Bayesian methods do not rely on classical assumptions. For the other three estimation methods, accuracy gradually improves as measurement relationships strengthen and structural relationships weaken. Notably, when R_s^2 increases from 0 to 0.4, keeping other factors constant, the RMSE of Bayesian + MLE decreases by approximately 12 decimal places. In contrast, the RMSE of joint ML and ML + ML decreases by 7 or 8 decimal places. This is somewhat surprising, as Bayesian-based estimates are generally expected to be more stable, given that the stationary distribution of a Markov chain is always the posterior distribution. In all settings, ML + ML tends to have slightly larger RMSE than joint ML, though the difference is minimal. Comparing Bayesian + ML with ML + ML estimation, the RMSEs are quite close when the sample size is large, the measurement relationship is strong, and the structural relationship is weak. In other scenarios, the RMSE of Bayesian + ML exceeds that of ML + ML approach. For example, in two extreme cases: $R_m^2 = 0.9, R_s^2 = 0, n = 1000$ and $R_m^2 = 0.5, R_s^2 = 0.8, n = 200$, the RMSE differences are on the order of 10^{-16} and 1, respectively.

The trend of RMSE changes is similar for the other two structural coefficients β_1 and β_3. It is noteworthy that β_2 represents the regression relationship of Y on M, but M itself includes the regression relationship on X represented by β_3. Therefore, it is hypothesized that the RMSE of β_2 should generally be slightly larger than that of β_3. However, a comparison of the RMSEs of β_2 and β_3 (see Table 13 in the Appendix) reveals that this difference is negligible.

Table 5: Root Mean Square Error (RMSE) of the structural parameter β_2 of two one-step methods and three two-step methods in 27 settings.

		one-step		two-step	
R_m^2	R_s^2	Joint ML	Joint Bayesian	ML + ML	Bayesian + ML
$n = 200$					
0.5	0.0	6.12×10^{-16}	1.55×10^{-3}	6.12×10^{-16}	3.14×10^{-11}
	0.4	3.71×10^{-9}	5.31×10^{-1}	4.53×10^{-8}	7.16×10^{-1}
	0.8	1.22×10^{-8}	7.32×10^{-1}	1.36×10^{-7}	1.05
0.7	0.0	6.31×10^{-16}	1.50×10^{-3}	6.31×10^{-16}	2.15×10^{-15}
	0.4	4.92×10^{-9}	3.14×10^{-4}	7.02×10^{-8}	6.39×10^{-16}
	0.8	9.64×10^{-9}	2.05×10^{-4}	1.77×10^{-7}	1.57×10^{-15}
0.9	0.0	4.41×10^{-16}	6.28×10^{-4}	4.41×10^{-16}	3.67×10^{-16}
	0.4	4.21×10^{-8}	1.36×10^{-3}	6.10×10^{-8}	2.89×10^{-2}
	0.8	1.68×10^{-8}	1.79×10^{-3}	8.09×10^{-8}	4.83×10^{-1}
$n = 600$					
0.5	0.0	5.89×10^{-16}	2.97×10^{-4}	5.90×10^{-16}	6.12×10^{-12}
	0.4	3.68×10^{-9}	4.98×10^{-3}	4.85×10^{-8}	1.94×10^{-2}
	0.8	1.30×10^{-8}	6.62×10^{-3}	1.35×10^{-7}	8.49×10^{-4}
0.7	0.0	5.48×10^{-16}	3.14×10^{-4}	5.48×10^{-16}	6.39×10^{-16}
	0.4	4.67×10^{-9}	1.73×10^{-3}	9.22×10^{-8}	1.88×10^{-2}
	0.8	9.69×10^{-9}	2.33×10^{-3}	1.80×10^{-7}	4.25×10^{-4}
0.9	0.0	4.75×10^{-16}	2.10×10^{-4}	4.75×10^{-16}	4.45×10^{-16}
	0.4	4.43×10^{-8}	5.47×10^{-4}	6.02×10^{-8}	1.58×10^{-4}
	0.8	1.74×10^{-8}	7.39×10^{-4}	7.50×10^{-8}	1.41×10^{-4}
$n = 1000$					
0.5	0.0	7.60×10^{-16}	2.58×10^{-4}	7.60×10^{-16}	7.80×10^{-16}
	0.4	3.87×10^{-9}	3.03×10^{-3}	4.92×10^{-8}	5.71×10^{-4}
	0.8	1.29×10^{-8}	3.98×10^{-3}	1.52×10^{-7}	5.58×10^{-4}
0.7	0.0	6.24×10^{-16}	2.05×10^{-4}	6.24×10^{-16}	1.57×10^{-15}
	0.4	4.75×10^{-9}	1.02×10^{-3}	1.64×10^{-7}	2.77×10^{-4}
	0.8	9.38×10^{-9}	1.42×10^{-3}	2.14×10^{-7}	3.00×10^{-4}
0.9	0.0	1.43×10^{-15}	1.79×10^{-4}	1.43×10^{-15}	1.75×10^{-15}
	0.4	3.98×10^{-8}	3.35×10^{-4}	5.81×10^{-8}	9.97×10^{-5}
	0.8	1.77×10^{-8}	4.57×10^{-4}	7.56×10^{-8}	1.09×10^{-4}

5.2 Estimated Standard Error Analysis

In the analysis of standard error estimation, the variance calculated by the five methods differs, so the results from all methods are included. This section focuses on the Simulation Standard Deviation

(s.d.) and its comparison with the Mean of Estimated Standard Error (m(se)). Taking β_1 as an example, Table 6 shows the s.d. and m(se) for two one-step methods. Both methods follow the pattern where higher measurement intensity and larger sample sizes result in smaller s.d. values, i.e., lower variability and higher precision of estimation. As the strength of the structural model decreases, the s.d. of joint ML tends to slightly decrease. When R_s^2 is small, the s.d. of joint ML is generally smaller than that of joint Bayesian estimation. Conversely, joint Bayesian shows greater precision when R_s^2 is larger. In the joint ML method, the s.d. exceeds the m(se) when structural intensity is high or medium and measurement intensity is low, i.e., the model underestimates the uncertainty or variability between samples. In contrast, for joint Bayesian estimation, the s.d. is generally smaller than the m(se), with the difference diminishing as sample size increases and measurement relationships strengthen. This discrepancy may be related to the choice of prior information.

Table 6: Simulation Standard Deviation vs Mean of Estimated Standard Error of β_1 in 27 settings for Joint ML and Joint Bayesian estimation methods.

			Joint MLE		Joint Bayesian Estimation	
n	R_m^2	R_s^2	s.d.	m(se)	s.d.	m(se)
200	0.5	0.0	0.08839	0.08839	0.15729	0.19348
		0.4	0.12059	0.09992	0.10411	0.21296
		0.8	0.14490	0.10497	0.10947	0.29597
	0.7	0.0	0.07829	0.07829	0.09084	0.13774
		0.4	0.09949	0.08326	0.08514	0.10290
		0.8	0.11643	0.08554	0.08740	0.12053
	0.9	0.0	0.07267	0.07267	0.07388	0.09442
		0.4	0.03250	0.03250	0.07519	0.08940
		0.8	0.10046	0.07458	0.07562	0.10221
600	0.5	0.0	0.05103	0.05103	0.06100	0.05200
		0.4	0.06962	0.05769	0.05846	0.07123
		0.8	0.08366	0.06060	0.06141	0.08583
	0.7	0.0	0.04520	0.04520	0.04555	0.04561
		0.4	0.05744	0.04807	0.04845	0.05804
		0.8	0.06722	0.04939	0.04969	0.06789
	0.9	0.0	0.04196	0.04196	0.04219	0.04223
		0.4	0.05067	0.04270	0.04293	0.05096
		0.8	0.05800	0.04306	0.04331	0.05837
1000	0.5	0.0	0.03953	0.03953	0.04846	0.03998
		0.4	0.05393	0.04468	0.04501	0.05465
		0.8	0.06480	0.04694	0.04731	0.06578
	0.7	0.0	0.03501	0.03501	0.03519	0.03524
		0.4	0.04449	0.03724	0.03742	0.04480

Continued on next page

Table 6: (continued)

			Joint MLE		Joint Bayesian Estimation	
n	R_m^2	R_s^2	s.d.	m(se)	s.d.	m(se)
		0.8	0.05207	0.03826	0.03842	0.05245
	0.9	0.0	0.03250	0.03250	0.03258	0.03262
		0.4	0.03925	0.03308	0.03316	0.03937
		0.8	0.04493	0.03335	0.03345	0.04509

Table 7 presents s.d. and m(se) for β_1 using three two-step methods. The s.d. and m(se) for the two-step methods are generally larger than those for the one-step methods. The trends of s.d. and its comparison with m(se) for the ML + ML and ML + draws + ML methods are very similar to those with joint ML. Specifically, in cases of high-intensity structural relationships and low-intensity measurement relationships, s.d. tends to be greater than m(se). Between ML + ML and its simplified version, ML + draws + ML, the s.d. and m(se) are very similar across all settings, demonstrating the efficiency of ML + draws + ML approach in variance estimation. The s.d. for the Bayesian + ML method tends to be larger than that of the other two-step methods when the sample size is small, especially when the measurement relationship is weak. However, with $n = 600, 1000$, its s.d. and m(se) become very similar to those of ML + ML, suggesting it can serve as an alternative to ML + ML under these conditions. In settings with small sample size and weak measurement relationships, Bayesian + ML estimation behaves similarly to joint Bayesian, with s.d. smaller than m(se). In other conditions, its behavior aligns more closely with joint ML estimation, where s.d. is slightly larger than m(se). The results for parameters β_2 and β_3 exhibit similar patterns.

Table 7: Simulation Standard Deviation vs Mean of Estimated Standard Error of β_1 in 27 settings for ML + ML, ML + draws + ML, Bayesian + ML estimation methods.

		ML + ML		ML + draws + ML		Bayesian + ML	
R_m^2	R_s^2	s.d.	m(se)	s.d.	m(se)	s.d.	m(se)
$n = 200$							
0.5	0.0	0.08839	0.08839	0.08869	0.08870	0.13504	0.23066
	0.4	0.12087	0.10010	0.11517	0.09454	0.27768	0.18257
	0.8	0.14540	0.10518	0.13855	0.09898	0.14053	0.23351
0.7	0.0	0.07829	0.07829	0.07829	0.07829	0.09667	0.07869
	0.4	0.09952	0.08328	0.09705	0.08089	0.12423	0.10123
	0.8	0.11643	0.08554	0.11348	0.08283	0.11378	0.08288
0.9	0.0	0.07267	0.07267	0.07268	0.07268	0.07276	0.07276
	0.4	0.08776	0.07397	0.08776	0.07397	0.08681	0.07302
	0.8	0.10047	0.07458	0.09968	0.07386	0.09909	0.08888

Continued on next page

Table 7: (continued)

		ML + ML		ML + draws + ML		Bayesian + ML	
R_m^2	R_s^2	s.d.	m(se)	s.d.	m(se)	s.d.	m(se)
$n = 600$							
0.5	0.0	0.05103	0.05103	0.05109	0.05109	0.05112	0.05112
	0.4	0.06979	0.05779	0.06641	0.05461	0.06674	0.05468
	0.8	0.08395	0.06073	0.07998	0.05728	0.08072	0.05752
0.7	0.0	0.04520	0.04520	0.04521	0.04521	0.04527	0.04528
	0.4	0.05746	0.04808	0.05603	0.04672	0.05604	0.04667
	0.8	0.06725	0.04940	0.06725	0.04940	0.06581	0.04798
0.9	0.0	0.04196	0.04196	0.04196	0.04196	0.04197	0.04197
	0.4	0.05067	0.04270	0.05031	0.04235	0.05034	0.04237
	0.8	0.05800	0.04306	0.05757	0.04266	0.05763	0.04268
$n = 1000$							
0.5	0.0	0.03953	0.03953	0.03956	0.03956	0.03964	0.03963
	0.4	0.05406	0.04477	0.05142	0.04230	0.05166	0.04241
	0.8	0.06503	0.04704	0.06195	0.04440	0.06228	0.04450
0.7	0.0	0.03501	0.03501	0.03501	0.03502	0.03504	0.03505
	0.4	0.04451	0.03725	0.04341	0.03619	0.04349	0.03624
	0.8	0.05210	0.03827	0.05210	0.03827	0.05090	0.03715
0.9	0.0	0.03250	0.03250	0.03250	0.03250	0.03251	0.03251
	0.4	0.03925	0.03308	0.03897	0.03281	0.03898	0.03282
	0.8	0.04493	0.03335	0.04460	0.03304	0.04462	0.03305

Next, the estimation of standard errors is further explored by calculating the proportion of times the 95% confidence or credible intervals for each approach covering the true parameter values. Encouragingly, except for some simulations under the Bayesian + ML method, the 95% intervals for all methods, including the "naive" standard error estimator in the global SAM, contain the true parameter values in each simulation. This differs from the conclusions of Kuha and Bakk (2024)'s application of latent variable models with categorical indicators, possibly because continuous variables are less traceable in estimation. Table 8 presents the coverage of the 95% credible interval for parameters β_1, β_2, β_3 under the Bayesian + ML estimation. The failure of the 95% confidence interval to cover the true value typically occurs when the sample size is small and the structural relationship intensity is high. Among all settings, the coverage for β_2 is generally the highest, while β_3 is the lowest. This is because β_2 includes part of the regression relationship represented by β_3, thereby reflecting its uncertainty in the standard deviation estimates.

Finally, there is an additional analysis concerning the estimated standard errors, specifically focusing on the "naive" estimator in the ML + ML (global SAM) method. Theoretically, using this estimate tends to underestimate the true standard errors for the parameters. The results of the simulation confirm

Table 8: Coverage of 95% confidence interval of β_1, β_2, β_3 for Bayes + ML estimation approach.

			Regression Coefficients		
R_m^2	R_s^2	n	β_1	β_2	β_3
0.5	0.4	200	97.78	98.87	92.56
		600	99.34	99.47	99.22
	0.8	200	94.37	95.21	92.34
		600	95.02	95.24	94.88
0.7	0.4	200	98.25	98.35	97.89
		600	99.01	99.03	99.00
	0.8	200	97.24	98.01	97.27
		600	97.24	98.01	97.27
0.9	0.4	200	98.89	98.82	98.01
	0.8	200	97.69	97.88	97.60

Note: The probability that the 95% confidence/credible intervals of parameters, for settings and methods not listed, cover the true values is 100%.

this: in every simulation of each setting, the "naive" estimator is consistently smaller than the accurate estimator. Table 9 lists the average percentage of the variance of the global SAM estimator accounted for by the "naive" estimator for the three structural model parameters in each setting. The portion of the variance explained by the "naive" estimator increases with stronger measurement intensity, weaker structural relationships, and larger sample sizes, though the latter effect is less pronounced. Specifically, each increase in R_m^2 results in a 3% to 5% increase in this proportion for some settings, while changing R_s^2 from 0.4 to 0 leads to an 8% increase in one setting. Among the three parameters, the "naive" estimator accounts for the smallest portion of the variance estimator for β_3 while the other two parameters are approximately similar.

Table 9: Simulation results for standard error estimates of estimated structural regression coefficients β_1, β_2, β_3: average % of the variance of SAM estimator accounted for by step-2 variance alone.

	$n = 200$			$n = 600$			$n = 1000$		
	β_1	β_2	β_3	β_1	β_2	β_3	β_1	β_2	β_3
$R_m^2 = 0.5$									
$R_s^2 = 0.0$	99.98	99.98	99.98	99.98	99.97	99.97	99.98	99.98	99.98
$R_s^2 = 0.4$	90.43	89.32	69.33	90.91	89.91	69.67	91.40	89.90	69.99
$R_s^2 = 0.8$	90.74	89.22	57.81	90.89	89.36	58.25	91.28	89.71	58.62
$R_m^2 = 0.7$									
$R_s^2 = 0.0$	99.99	99.99	99.99	99.99	99.99	99.99	99.99	99.99	99.99
$R_s^2 = 0.4$	95.12	94.47	82.82	95.48	95.07	83.06	95.92	94.92	83.45
$R_s^2 = 0.8$	95.10	94.07	73.55	95.27	94.65	74.08	95.88	94.94	74.92
$R_m^2 = 0.9$									

Continued on next page

Table 9: (continued)

	$n = 200$			$n = 600$			$n = 1000$		
	β_1	β_2	β_3	β_1	β_2	β_3	β_1	β_2	β_3
$R_s^2 = 0.0$	99.99	99.99	99.99	99.99	99.99	99.99	99.99	99.99	99.99
$R_s^2 = 0.4$	98.59	98.38	94.67	99.01	98.87	95.27	99.21	99.14	95.52
$R_s^2 = 0.8$	98.54	98.18	90.83	98.95	98.65	91.44	99.23	98.87	91.93

5.3 Distribution of Measurement Parameters Analysis

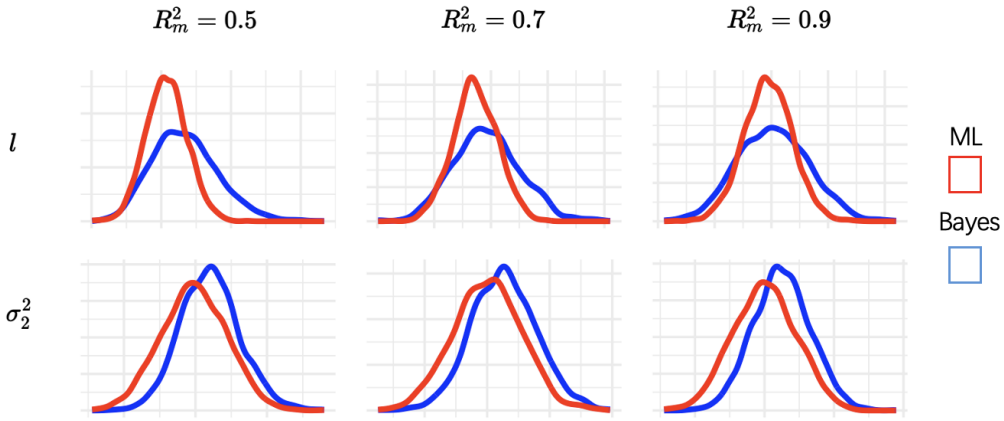
Previous sections indicate that estimates under ML-based and Bayesian frameworks differs due to the incorporation of prior information in the Bayesian approach. This leads to different parameter distributions, which affects not only the parameter estimation results themselves but also sampling process of the measurement parameters between the first and second steps in ML + draws + ML and Bayesian + ML methods. These parameters are divided into two components: loadings and measurement error variance. For loadings, the posterior distribution in both methods is normal, differing only in mean and variance. However, for measurement error variance, the posterior distributions differ entirely: in the ML + draws + ML method, they are assumed to be normally distributed, whereas in the Bayesian + ML method, they follow an inverse gamma distribution. This section will explore the differences in the distribution states of the measurement parameters under different settings through graphical representations.

In the first step, loadings and measurement error variance are estimated and sampled. For each setting, the true values of the free loadings are the same, denoted as l, and the true values of the measurement error variance are of two types: σ_1^2 and σ_2^2. Among the three parameters, l and σ_2^2 are selected to explore the difference in their sampling distributions under ML + draws + ML and ML + Bayesian estimation across various settings. In each setting, 2,000 datasets are randomly generated. Since summarising the sampling distribution of all datasets is challenging, multiple datasets are studied and a typical dataset is presented as an example for each setting.

Under the prior distribution set in the simulation experiment, the estimated mean of l from ML method is slightly smaller than that of the Bayesian estimation, but the estimated variance is much larger than the Bayesian estimation results. The posterior means of σ_1^2 and σ_2^2 are also slightly smaller than ML estimation compared with Bayesian approach, but the variance differences are minimal. Next, by fixing two conditions in one setting, explore how changes in another condition affect the distribution of l and σ_2^2.

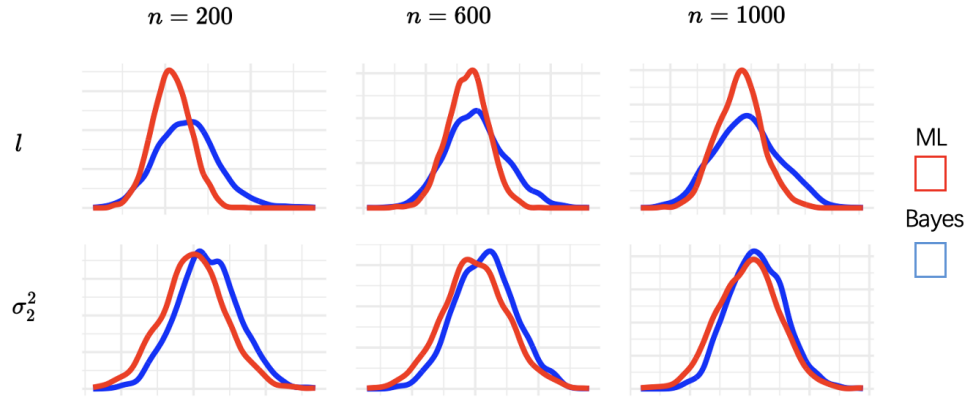
As shown in Figure 3, with R_s^2 fixed at 0 and the sample size n fixed at 200, the central tendency of the estimated distribution of l under both methods becomes closer as the strength of the measurement model's relationships increases. However, the variance does not change significantly. The trend for σ_2^2 is similar. The posterior distribution of σ_2^2 should follow an inverse-gamma distribution, but its shape

Figure 3: Density plots of posterior draws of l, σ_2^2 for ML and Bayesian estimation with different R_m^2, controlling for $R_s^2 = 0, n = 200$.



is quite similar to the ML results. This phenomenon might be due to the large shape parameter in the posterior probability density under the given priors in the simulation, making the inverse-gamma distribution visually resemble a normal distribution.

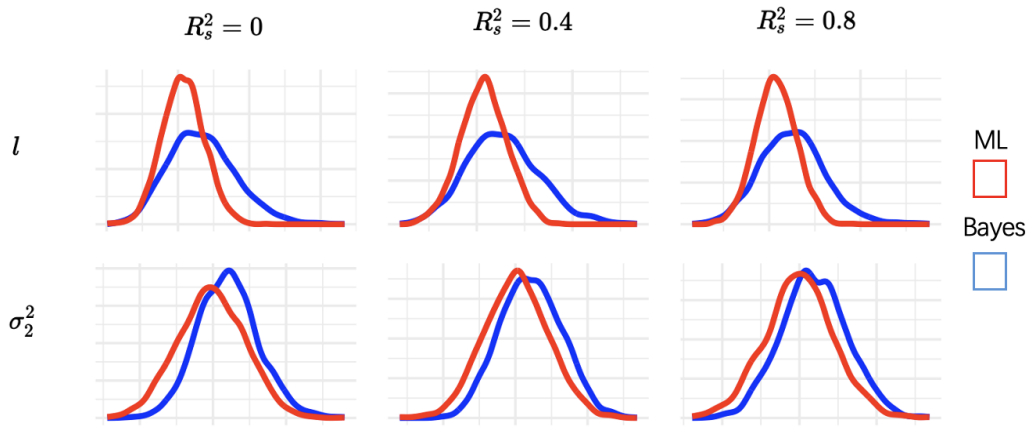
Figure 4: Density plots of posterior draws of l, σ_2^2 for ML and Bayesian estimation with different n, controlling for $R_m^2 = 0.5, R_s^2 = 0.8$.



As n increases, Figure 4 shows the changes in l and σ_2^2, with R^2 fixed at 0.5 for the measurement model and 0.8 for the structural model. It is evident that as the sample size increases, the estimated mean, variance and distribution shape of two parameters under Bayesian estimation become increasingly similar to those under ML estimation. This effect is the most pronounced for l when the sample size increases from 200 to 600. This occurs because the dataset generation follows the classical normality assumptions, which reduce the influence of priors as the sample size grows.

The first step refers to the estimation of the measurement model. Therefore, changes in the strength

Figure 5: Density plots of posterior draws of l, σ_2^2 for ML and Bayesian estimation with different R_s^2, controlling for $R_m^2 = 0.5, n = 200$.



of the structural model R_s^2 do not affect the estimation in the first step. However, when R_s^2 is used to generate the dataset, it may influence the measurement model data by affecting the relationships between latent variables. This effect is negligible, as shown by parameter distributions of multiple datasets (see the example in Figure 5).

6 Empirical Study

This section conducts an empirical study using a real-world dataset to explore the proposed methods. Apart from focusing on the estimated means and standard errors, this section includes two additional aspects not covered in Section 5: the possibility of interpretational confounding and the computational complexity. Since the length of a Markov chain to reach its stationary distribution varies with different datasets and initial values, the joint ML and SAM methods are selected to represent the one-step and two-step SEM approaches respectively for a comparative study.

The data employed in this study originates from Round 10 of the European Social Survey (ESS) (European Social Survey European Research Infrastructure Consortium, 2023a, 2023b), conducted between September 2020 and August 2022. This biennial cross-national survey monitors and interprets changing public attitudes and values within Europe, with Round 10 specially focusing on democracy and the use of digital communication. A total of 59,685 respondents from 31 countries participated in Round 10, with 9 countries employing self-administered methods (web and paper) and 22 countries using face-to-face fieldwork approaches (in-person and video interviews).

A SEM concerning digital social contacts and work-life balance is created based on the ESS Topline Results Series 12 (Abendroth et al., 2023). The model includes three latent factors: costsDSC (the costs associated with digital social contacts), benefDSC (the benefits derived from digital social contacts), wlconDSC (the work-life conflict resulting from digital social contacts), along with 9 indicators explained in Table 10. In the measurement submodel, costsDSC is measured by mcpriv, mcmsinf and mcinter; benefDSC by mccoord, mcclose and mcwrhom; wlconDSC by pfmfdjba, jbprtfp and trdawrk. Notably, mcpriv, mccoord and pfmfdjba serves as anchor indicators. Some demographic information is also considered for inclusion in the structural model, while in the basic version of the structural submodel, wlconDSC is influenced by both costsDSC and benefDSC, with a correlation assumed between costsDSC and benefDSC. The values of nine items are ordinal integers, but for convenience, all indicators and latent factors are treated as continuous variables. These item values have been adjusted using post-stratification weights and design weights to eliminate the effects of population characteristics.

Table 10: The interpretation of indicators.

Indicator	Interpretation
mcpriv	Online/mobile communication undermines personal privacy
mcmsinf	Online/mobile communication exposes people to misinformation
mcinter	Online/mobile communication makes work and personal life interrupt each other
mccoord	Online/mobile communication makes it easy to coordinate and manage activities
mcclose	Online/mobile communication makes people feel closer to one another
mcwrkhom	Online/mobile communication makes it easy to work from home or place of choice
pfmfdjba	Partner/family fed up with pressure of your job, how often

Continued on next page

Table 10: Indicator Explanation (continued)

Indicator	Interpretation
jbprtfp	Job prevents you from giving time to partner/family, how often
trdawrk	Too tired after work to enjoy things like doing at home, how often

Two analyses are carried out with the empirical example. The first analysis is based on the data from a single country, Germany. In this analysis, the measurement model remains constant while the structural model is altered to explore interpretational confounding and preliminarily examine the model structure. The second analysis is a cross-sectional study of data from all 31 countries. In this analysis, parameter values in the structural model vary by country, allowing an investigation of the issues of computational complexity. It is observed that, unlike in a simulation study, the dataset contains some missing values. Assume that these missing values are missing completely at random.

6.1 One Country Research: Germany

After data cleaning and preprocessing, the German sample emerges as the largest, with 8,725 valid entries among the 31 countries. To avoid biases caused by insufficient data entries, the German data is used for preliminary exploration. Several demographic characteristics are added to the structural model to directly influence wlconDSC: agea (age of respondents, 15-90), hhlio12 (child aged 12 or over lives in the same household, 0-1), hhlipnt (parent lives in the same household, 0-1), wkhtot (total hours normally worked per week in main job, including overtime, 0-168). Consequently, four types of structural models are established:

model 1. only the three latent factors;
model 2. the three latent factors + agea;
model 3. the three latent factors + agea + hhlio12 + hhlipnt;
model 4. the three latent factors + agea + hhlio12 + hhlipnt + wkhtot.

The four models are fitted using both joint ML and SAM approaches, with all loading estimates statistically significant at $\alpha = 0.01$. The estimated loadings are presented in Table 11. The first column displays results from the SAM method, where estimated loading values are identical across all four models. The subsequent four columns present the loading estimates of the four models obtained via joint ML method. In SAM, the structural model is estimated first, without considering correlations between factors at this stage. In contrast, in the one-step method, the structural part and the measurement part are estimated simultaneously to maximise the likelihood function. Thus, when employing the SAM approach, the parameter estimates of the measurement part are unaffected by the structural part, avoiding interpretational confounding. Conversely, in the joint ML method, modifications in the structural part, such as adding or removing variables or altering relationships, also influence the parameter estimates of the measurement part. Although these changes are relatively minor, they introduce some discrepancies. For example, among the three indicators associated with benefDSC, results from model 1 indicate that mccoord has the strongest measurement relationship

with the latent factor, whereas the other three models show mcclose as the indicator with the highest loading.

Table 11: Estimated loading values of 4 models via SAM and joint ML methods.

model	**SAM**	**1-step model 1**	**1-step model 2**	**1-step model 3**	**1-step model 4**
mcpriv	1.000				
mcmsinf	0.571	0.600	0.598	0.599	0.596
mcinter	0.635	0.673	0.670	0.671	0.668
mccoord	1.000				
mcclose	1.287	0.992	1.019	1.015	1.000
mcwrkhom	1.055	0.901	0.922	0.920	0.918
pfmfdjba	1.000				
jbprtfp	1.204	1.184	1.195	1.204	1.245
trdawrk	0.661	0.662	0.667	0.666	0.673

Regarding the structural model parameter estimates, the significant tests for the individual parameters generally perform worse than those in the measurement model, with some parameters showing p-values exceeding 0.05. The estimation results from both approches are quite similar. Specifically, costsDSC has a positive effect on wlconDSC, while benefDSC appears to mitigate it. Younger people experience slightly less work-life conflict from digital social contacts than older individuals. Additionally, respondents who live apart from their children over 12 or parents find it more difficult to achieve work-life balance. Notably, there exists a very slight positive correlation between costsDSC and benefDSC, suggesting that, for most individuals, digital social contacts can be a double sword, offering benefits while also present challenges.

By computing the R^2 for each equation within the model and comparing it with the simulation part in Section 5, it is found that in the measurement model, there is low or moderate measurement strength between each indicator and its corresponding latent factor. In the structural model, the regression relationship strength between wlconDSC and other variables is low. By comparing the Akaike Information Criterion(AIC) and Bayesian Information Criterion(BIC), in both joint ML and SAM methods, it is found that model 1, where the structural model involves only latent variables, has the lowest AIC and BIC values. Therefore, model 1 will be employed in the subsequent cross-national analysis.

6.2 Cross-national Research

Information from all 31 countries is utilized in this section to construct a cross-national structural equation model. The model framework follows the one with the lowest AIC and BIC identified in Section 6.1, which involves only three latent variables in the structural part. Previous literature (Abendroth et al., 2023) indicates that the ordinary least square regression between wlconDSC and one of the forms of online/mobile communication, the frequency of written digital communication with the line manager, varies across different countries. Although there is an overall positive relationship between the two variables, the regression effect is more pronounced in Israel, North Macedonia, and

Sweden, whereas in France and Greece, there is almost no significant relationship. When exploring the perceived opportunities and risks of digital social contacts, a positive correlation is observed between "Makes work and life interrupt each other" (mcinter) and "Makes work and life interrupt each other" (mccoord) in all countries, though the specific strength of this correlation varies. Therefore, it is assumed the measurement model parameters, such as intercepts, loading and measurement error variance, are identical across all countries. However, structural model parameters, like correlation and regression coefficients, are allowed to vary by country. This approach ensures consistent measurement of individual values and behaviors across all countries, while allowing for the exploration of cross-nation and regional differences at a macro level.

In the cross-national research, the dataset used in step 1 (59,685 observations) is substantially larger than the dataset employed in step 2 (typically ranging between 1,000 and 2,000 observations). As a result, the uncertainty associated with the parameters from step 1 essentially contributes negligibly to the standard errors of the estimated structural coefficients. In addition, as mentioned previously, the procedure for SAM method computing standard errors of structural parameters is complicated. Therefore, the ML + ML without draws method, which treats the parameters from step 1 as known, is selected for the two-step approach to conduct the experiment.

The ML + ML without draws method is concurrently applied to the non-cross-national scenario, assuming equal parameter estimates for both the measurement and structural models across all countries. In this context, the maximum log-likelihood value is lower than that of the cross-nation model, supporting the validity of analysing models on a country-by-country basis. In the two-step estimation approach for cross-national research, the measurement model is first estimated using data from all countries. In the next step, with measurement model parameters held constant, the full model is fitted, allowing the structural parameters to vary across countries. This procedure yields distinct correlation and regression coefficients for each country, as depicted in Table 12. The two regression coefficients estimates vary minimally across countries. costsDSC and benefDSC imparct wlconDSC with promoting and diminishing effects, respectively, both relatively small in magnitude. Belgium and Austria show the largest effects of costsDSC on wlconDSC, while the United Kingdom and the Netherlands exhibit relatively larger effects of benefDSC on wlconDSC. In Greece and the Netherlands, individuals generally do not perceive that the costs of digital social contacts worsen the work-life imbalance. This suggests that such costs may be more pronounced in other areas rather than family or work. In Poland, individuals also believe that the digital social contacts' conveniences do not significantly improve work-life balance, indicating that these conveniences may have limited relevance to family and work dynamics. Covariance estimates between costsDSC and benefDSC vary substantially, though most are positive. In Montenegro, the correlation between these two variables is the strongest. However, in Bulgaria and Australia, costsDSC and benefDSC show a negative correlation, suggesting that these costs and benefits may be complementary in these regions.

Table 12: Estimates for structural parameters of 31 Countries.

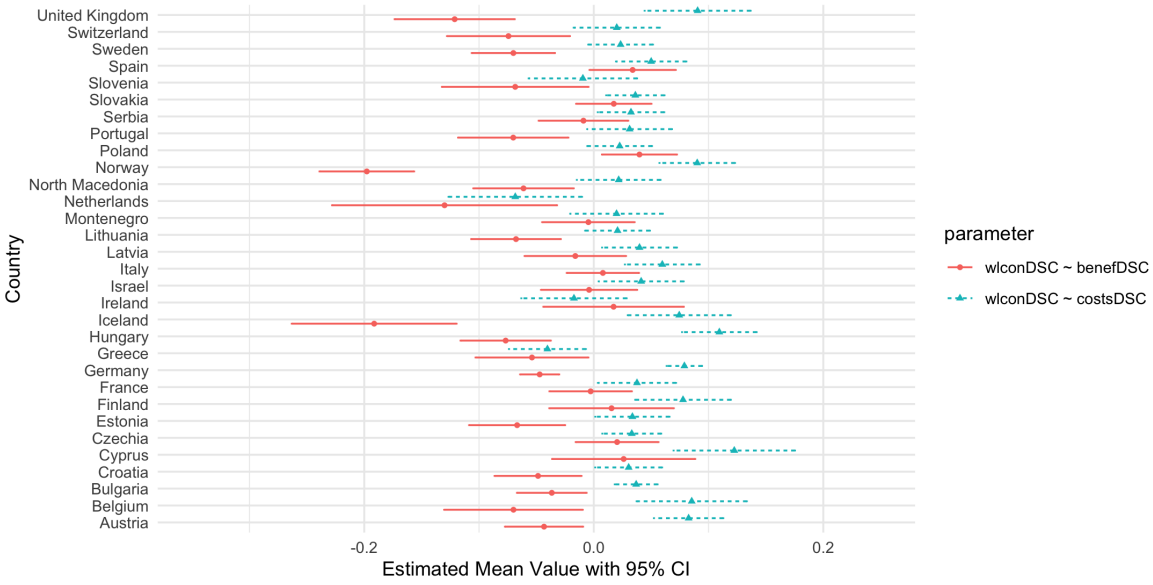
Country	wlconDSC~costsDSC	wlconDSC~benefDSC	costsDSC~~benefDSC
Belgium	0.085	-0.070	1.018
Bulgaria	0.037	-0.037	-0.442
Switzerland	0.020	-0.074	0.874
Czechia	0.033	0.020	0.378
Estonia	0.034	-0.067	0.498
Finland	0.078	0.015	0.886
France	0.038	-0.003	0.190
United Kingdom	0.090	-0.121	1.032
Greece	-0.040	-0.054	0.375
Croatia	0.030	-0.049	0.415
Hungary	0.109	-0.077	0.447
Ireland	-0.017	0.017	0.973
Iceland	0.074	-0.191	0.783
Italy	0.060	0.008	0.986
Lithuania	0.021	-0.068	-0.201
Montenegro	0.020	-0.005	3.133
North Macedonia	0.022	-0.061	0.592
Netherlands	-0.068	-0.130	0.796
Norway	0.090	-0.198	0.354
Portugal	0.031	-0.070	1.061
Slovenia	-0.009	-0.068	1.495
Slovakia	0.036	0.017	0.163
Austria	0.083	-0.043	-0.384
Cyprus	0.122	0.026	1.178
Germany	0.079	-0.047	0.274
Spain	0.050	0.034	0.922
Israel	0.041	-0.004	1.144
Latvia	0.040	-0.016	0.919
Poland	0.023	0.040	0.718
Serbia	0.032	-0.009	0.215
Sweden	0.023	-0.070	-0.026

Note: The estimates highlighted with gray shading are not statistically significant at the 5% significant level. All values are rounded to three decimal places.

Figure 6 illustrates the 95% confidence intervals for the effects of benefDSC and costsDSC on wlconDSC in each country. These intervals are derived from the estimated mean values and standard errors of the regression coefficients in the structural model. For each country, the confidence intervals for the two parameters are relatively similar in length, indicating that individuals' attitudes toward

online communication, whether positive or negative, are similarly clustered or dispersed with no significant variation. Across all countries, respondents in Netherlands exhibit relatively dispersed views, whereas attitudes are more unified among respondents in Germany and Bulgaria. The 95% confidence intervals for the covariance between costsDSC and benefDSC, shown in Figure 8 in the Appendix, reveal that views in Montenegro are quite scattered, while those in the Netherlands, Greece and Germany are more consistent. The concentrated estimates for the three parameters in Germans may result from the influence of German societal and cultural factors, as well as the larger sample size compared to other countries.

Figure 6: Estimated mean values and 95% confidence intervals of regression coefficients of 31 countries.

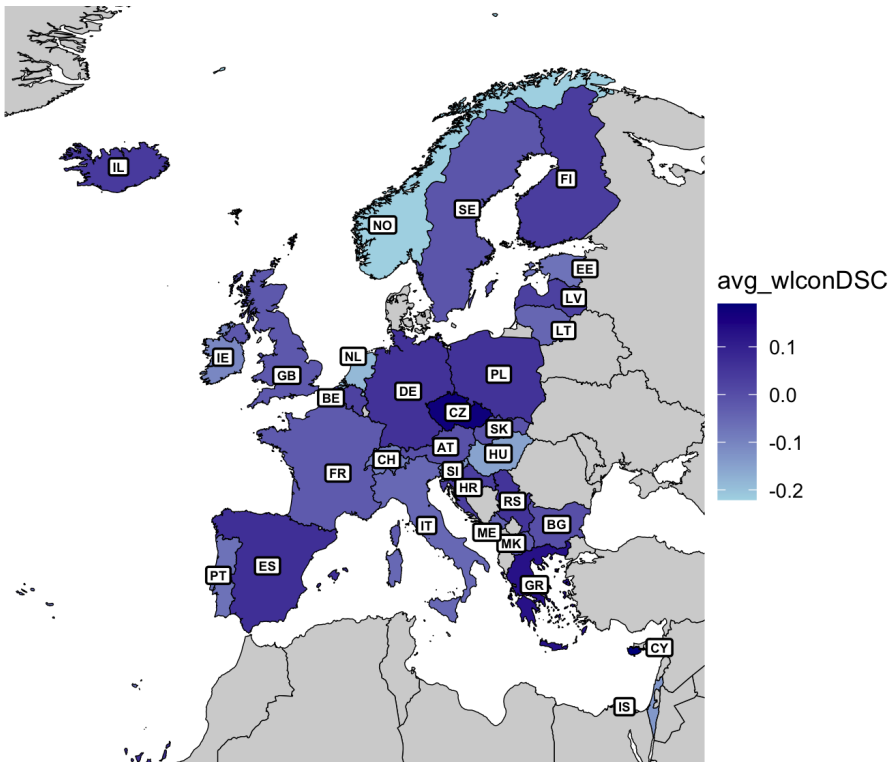


Considering work-life balance factors like labor policies and electronic communications development, which may be influenced by geographical factors (i.e., neighboring countries may exhibit higher correlations), the average wlconDSC values for each country are predicted and mapped in Figure 7. In Czechia, individuals experience the most pronounced conflict between work and personal life due to digital social contacts. Surrounding countries also struggle to varying degrees with work-life balance. Notably, Israel, the only country geographically located in Asia in this study, also faces significant issues with managing work-life conflicts due to online communication. Among surveyed European countries, respondents from border countries the Netherlands, Norway and Iceland generally manage to balance work and life relatively well. This suggests that technological and labor interactions among European countries are frequent, especially between neighboring countries. Thus, the work-life balance issues discussed here are indeed influenced by spatial factors.

The joint ML estimation is also applied to fit the model to compare computational complexity. The model comprises 186 parameters, with all indicators and latent variables being continuous. When using joint estimation, both measure and structural models are fitted simultaneously with data from all countries, resulting in a total computational time of 12.96 minutes. In contrast, with the two-step ML + ML approach, the first step utilizes the entire dataset, while the second step can be viewed as being performed separately for each country. This approach results in a total computational time of

1.16 minutes. When some of indicators and latent factors are categorical, the likelihood function is non-closed, leading to a significantly greater difference in computation time between two methods (Kuha and Bakk, 2024).

Figure 7: The heat map of average wlconDSCs level of 31 countries.



Note: The gradient from light blue to dark blue represents the average wlconDSC level ranging from low to high. Gray areas indicate countries that are not included in the survey pool.

7 Conclusion and Discussion

This research explores the application of a two-step approach framework to SEMs, where all indicators and latent factors are continuous variables. The study highlights several key properties and performance characteristics of this approach. Compared with one-step method, the two-step method offers certain practical and conceptual advantages. However it may not always yield results as accurate as the one-step method. The study includes two applications. The simulation study examines the accuracy of certain versions of two-step estimation under different settings. The empirical study demonstrates the advantages of the two-step approach over the one-step method using a specific example. The main findings and ideas of further research are discussed below.

7.1 Key Findings

Overall, the results indicate that larger sample sizes, increased certainty in the measurement relationships, and increased uncertainty in the structural relationships tend to yield the least biased estimates of the structural model parameters, termed the "ideal setting". When using the SAM method, the accuracy of the structural model estimates is nearly identical to those obtained from the joint ML method, except in extreme cases (e.g., when the strength of the structural relationships is very high). Applying the ML + draws + ML estimation with 2,000 samples of measurement model parameters can also achieve high precision in structural model estimation. Under Bayesian + ML estimation with weakly informative priors produces highly accurate point estimates of the structural model parameters, except when the measurement model's strength is extremely low. In these corresponding situations, Bayesian + ML or ML + draws + ML estimation can serve as alternatives to SAM. The study also found that joint Bayesian estimation has a notable advantage over joint ML method: it is less influenced by sample size, resulting in lower variability in outcomes.

In the empirical example of this study, the measurement model parameters are assumed to be uniform, while the structural model parameters are estimated by group. In this scenario, the first step uses much more data than the second for each group, making the two-step estimation method significantly advantageous in terms of computational complexity. In the empirical example, the one-step method takes more than ten times longer than the two-step method, and a disparity that would be even more pronounced in more intractable model forms, such as latent trait models (Kuha and Bakk, 2024). The example also confirmed the issue of interpretational confounding with the one-step estimation, where changes in the structural model can alter the order of strength with which indicators measure latent factors. Moreover, this example illustrates an application of SEM and two-step analysis in social sciences.

7.2 Further Research Directions

The limitation and potential extensions of this study focus on three key areas.

Firstly, the simulation study does not include covariates $\boldsymbol{Z_{ip}}$ or response variables $\boldsymbol{Z_{io}}$ in the model setting. These variables can take various forms and distributions, adding complexity to the model and enriching the comparison between one-step and two-step estimation methods. Bakk and Kuha (2018) introduced a uniformly distributed $\boldsymbol{Z_{ip}}$ and a normally distributed $\boldsymbol{Z_{io}}$ into two model structures in a simulation study and found that the two-step estimation performed significantly worse than the one-step method under non-ideal conditions. Future research could simulate more diverse data types and complex relationships in latent variable models to further explore the properties of the two-step estimation method.

Secondly, the Bayesian methods in the second step, as mentioned in Section 4.2.2, are not applied in the subsequent analysis due to efficiency considerations. However, during the simulation study, Bayesian estimation proves to be highly stable and less dependent on large sample sizes. Furthermore, in moderate scenarios, Bayesian + ML method results in highly accurate structural model estimates. Therefore, employing Bayesian estimation in both steps might combine the advantages of convenient sampling, stable estimation, and high precision, potentially outperforming all other estimation method combinations under certain settings. This direction could be further explored.

Finally, regarding the application of Bayesian methods, this study uses weakly informative priors. Due to the uncontrolled convergence time of Markov chains, Bayesian estimation is also not used to compare computationally complexity in the empirical study. Actually there remains significant potential for exploring Bayesian methods in social sciences and latent variable models. Moreover, research on Bayesian-based two-step methods currently seems to be very limited, presenting an opportunity for future research expansion in this direction.

Bibliography

Abendroth, A.-K., Lükemann, L., Hargittai, E., Billari, F., Treas, J., & van der Lippe, T. (2023). Digital social contacts in work and family life: Topline results from round 10 of the european social survey. https://www.europeansocialsurvey.org/sites/default/files/2023-11/TL012%20DSC%20FINAL.pdf

Anderson, J. C., & Gerbing, D. W. (1988). Structural equation modeling in practice: A review and recommended two-step approach. *Psychological Bulletin*, *103*(3), pp411–423. https://doi.org/10.1037/0033-2909.103.3.411

Arminger, G., & Muthén, B. O. (1998). A bayesian approach to nonlinear latent variable models using the gibbs sampler and the metropolis-hastings algorithm. *Psychometrika*, *63*, pp271–300. https://doi.org/10.1007/BF02294856

Bakk, Z., & Kuha, J. (2018). Two-step estimation of models between latent classes and external variables. *Psychometrika*, *83*(4), pp871–892. https://doi.org/10.1007/s11336-017-9592-7

Bakk, Z., Tekle, F. B., & Vermunt, J. K. (2013). Estimating the association between latent class membership and external variables using bias-adjusted three-step approaches. *Sociological Methodology*, *43*(1), pp272–311. https://doi.org/10.1177/0081175012470644

Bandeen-Roche, K., Miglioretti, D. L., L. Zeger, S., & Rathouz, P. J. (1997). Latent variable regression for multiple discrete outcomes. *Journal of the American Statistical Association*, *92*(440), pp1375–1386. https://doi.org/10.1080/01621459.1997.10473658

Bolck, A., Croon, M., & Hagenaars, J. (2004). Estimating latent structure models with categorical variables: One-step versus three-step estimators. *Political Analysis*, *12*(1), pp3–27. https://doi.org/10.1093/pan/mph001

Bollen, K. A. (1989). *Structural equations with latent variables*. John Wiley & Sons Inc. https://doi.org/10.1002/9781118619179

Bollen, K. A. (1996). An alternative two stage least squares (2sls) estimator for latent variable equations. *Psychometrika*, *61*(1), pp109–121. https://doi.org/10.1007/BF02296961

Bollen, K. A. (2000). Modeling strategies: In search of the holy grail. *Structural Equation Modeling: A Multidisciplinary Journal*, *7*(1), pp74–81. https://doi.org/10.1207/S15328007SEM0701_03

Browne, M. W., & Arminger, G. (1995). Specification and Estimation of Mean- and Covariance-Structure Models. In G. Arminger, C. C. Clogg & M. E. Sobel (Eds.), *Handbook of Statistical Modeling for the Social and Behavioral Sciences*. Plenum Press. https://doi.org/10.1007/978-1-4899-1292-3_4

Burt, R. S. (1973). Confirmatory factor-analytic structures and the theory construction process. *Sociological Methods & Research*, *2*(2), pp131–190. https://doi.org/10.1177/004912417300200201

Burt, R. S. (1976). Interpretational confounding of unobserved variables in structural equation models. *Sociological Methods & Research*, *5*(1), pp3–52. https://doi.org/10.1177/004912417600500101

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., & Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, *76*(1), pp1–32. https://doi.org/10.18637/jss.v076.i01

Dayton, C. M., & Macready, G. B. (1988). Concomitant-variable latent-class models. *Journal of the American Statistical Association*, *83*(401), pp173–178. https://doi.org/10.1080/01621459.1988.10478584

Depaoli, S. (2021). *Bayesian Structural Equation Modeling*. Guilford Publications.

Devlieger, I., Mayer, A., & Rosseel, Y. (2016). Hypothesis testing using factor score regression: A comparison of four methods. *Educational and Psychological Measurement*, *76*(5), pp741–770. https://doi.org/10.1177/0013164415607618

Embretson (Whitely), S. (1984). A general latent trait model for response processes. *Psychometrika*, *49*(2), pp175–186. https://doi.org/10.1007/BF02294171

European Social Survey European Research Infrastructure Consortium. (2023a). ESS10 integrated file, edition 3.2 [Data set]. https://doi.org/10.21338/ess10e03_2

European Social Survey European Research Infrastructure Consortium. (2023b). ESS10 Self-completion - integrated file, edition 3.1 [Data set]. https://doi.org/10.21338/ess10sce03_1

Ferron, J. M., & Hess, M. R. (2007). Estimation in sem: A concrete example. *Journal of Educational and Behavioral Statistics*, *32*(1), pp110–120. https://doi.org/10.3102/1076998606298025

Gelman, A., Carlin, J. B., Stern, H. S., & Rubin, D. B. (1995). *Bayesian Data Analysis*. Chapman & Hall Ltd. https://doi.org/10.1201/9780429258411

Goldstein, H., & Browne, W. J. (2002). Multilevel factor analysis modelling using Markov chain Monte Carlo (MCMC) estimation. In G. A. Marcoulides & I. Moustaki (Eds.), *Latent Variable and Latent Structure Models*. Psychology Press.

Gong, G., & Samaniego, F. J. (1981). Pseudo maximum likelihood estimation: Theory and applications. *The Annals of Statistics*, *9*(4), pp861–869. https://doi.org/10.1214/aos/1176345526

Gourieroux, C., & Monfort, A. (1995). *Statistics and Econometric Models* (Q. Vuong, Ed.). Cambridge University Press. https://doi.org/10.1017/CBO9780511751950

Jöreskog, K. G. (1970). A general method for estimating a linear structural equation system. *ETS Research Bulletin Series*, *1970*(2), ppi–41. https://doi.org/10.1002/j.2333-8504.1970.tb00783.x

Jöreskog, K. G., & Sörbom, D. (1986). *LISREL VI: Analysis of Linear Structural Relationships by Maximum Likelihood and Least Squares Methods* (3rd). Scientific Software.

Kaplan, D. (2009). *Structural Equation Modeling: Foundations and Extensions* (2nd). SAGE Publications, Inc. https://doi.org/10.4135/9781452226576

Kaplan, D., & Depaoli, S. (2013). Bayesian Statistical Methods. In T. D. Little (Ed.), *The Oxford Handbook of Quantitative Methods*. Oxford University Press.

Koopmans, T. C., Rubin, H., & Leipnik, R. B. (1950). Measuring the Equation Systems of Dynamic Economics. In T. C. Koopmans (Ed.), *Statistical Inference in Dynamic Economic Models*. Wiley.

Kuha, J., & Bakk, Z. (2024). Two-step estimation of latent trait models. https://arxiv.org/abs/2303.16101

Lance, C. E., Cornwell, J. M., & Stanley, A. M. (1988). Limited information parameter estimates for latent or mixed manifest and latent variable models. *Multivariate Behavioral Research*, *23*(2), pp171–187. https://doi.org/10.1207/s15327906mbr2302_3

Lee, S.-Y. (2007). *Structural Equation Modeling: A Bayesian Approach*. John Wiley & Son Ltd. https://doi.org/10.1002/9780470024737

Lee, S.-Y., & Shi, J.-Q. (2000). Bayesian analysis of structural equation model with fixed covariates. *Structural Equation Modeling: A Multidisciplinary Journal*, *7*(3), pp411–430. https://doi.org/10.1207/S15328007SEM0703_3

Levy, R. (2017). Distinguishing outcomes from indicators via bayesian modeling. *Psychological Methods*, *22*(4), pp632–648. https://doi.org/10.1037/met0000114

Levy, R. (2023). Precluding interpretational confounding in factor analysis with a covariate or outcome via measurement and uncertainty preserving parametric modeling. *Structural Equation Modeling: A Multidisciplinary Journal*, *30*(5), pp719–736. https://doi.org/10.1080/10705511.2022.2154214

Lindley, D. V., & Smith, A. F. M. (1972). Bayes estimates for the linear model. *Journal of the Royal Statistical Society, Series B (Methodological)*, *34*(1), pp1–18. https://doi.org/10.1111/j.2517-6161.1972.tb00885.x

Lu, I. R. R., & Thomas, D. R. (2008). Avoiding and correcting bias in score-based latent variable regression with discrete manifest items. *Structural Equation Modeling : A Multidisciplinary Journal*, *15*(3), pp462–490. https://doi.org/10.1080/10705510802154323

Martin, J. K., & McDonald, R. P. (1975). Bayesian estimation in unrestricted factor analysis: A treatment for heywood cases. *Psychometrika*, *40*, pp505–517. https://doi.org/10.1007/BF02291552

Merkle, E. C., Fitzsimmons, E., Uanhoro, J., & Goodrich, B. (2021). Efficient bayesian structural equation modeling in stan. *Journal of Statistical Software*, *100*(6), pp1–22. https://doi.org/10.18637/jss.v100.i06

Merkle, E. C., & Rosseel, Y. (2018). Blavaan: Bayesian structural equation models via parameter expansion. *Journal of Statistical Software*, *85*(4), pp1–30. https://doi.org/10.18637/jss.v085.i04

Muthén, B. (2010). *Bayesian analysis in mplus: A brief introduction* (tech. rep.) (Technical Report, Version 3). Mplus. https://api.semanticscholar.org/CorpusID:53516449

Olsson, U. H., Foss, T., Troye, S. V., & Howell, R. D. (2000). The performance of ml, gls, and wls estimation in sem under conditions of misspecification and non-normality. *Structural Equation Modeling: A Multidisciplinary Journal*, *7*(4), pp557–595. https://doi.org/10.1207/S15328007SEM0704_3

Rosseel, Y. (2012). Lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, *48*(2), pp1–36. https://doi.org/10.18637/jss.v048.i02

Rosseel, Y., & Loh, W. (2022). A structural after measurement approach to structural equation modeling [Advance online publication]. *Psychological Methods*. https://doi.org/10.1037/met0000503

RStudio Team. (2021). *Rstudio: Integrated development environment for R*. RStudio, PBC. Boston, MA. https://www.rstudio.com/

Scheines, R., Hoijtink, H., & Boomsma, A. (1999). Bayesian estimation and testing of structural equation models. *Psychometrika*, *64*(1), pp37–52. https://doi.org/10.1007/BF02294318

Skrondal, A., & Kuha, J. (2012). Improved regression calibration. *Psychometrika*, *77*(4), pp649–669. https://doi.org/10.1007/s11336-012-9285-1

Skrondal, A., & Laake, P. (2001). Regression among factor scores. *Psychometrika*, *66*(4), pp563–575. https://doi.org/10.1007/BF02296196

Spearman, C. (1904). "General intelligence," objectively determined and measured. *The American Journal of Psychology*, *15*(2), pp201–292. http://www.jstor.org/stable/1412107

Thurstone, L. L. (1947). *Multiple Factor Analysis: A Development and Expansion of Vectors of the Mind*. University of Chicago Press.

Vermunt, J. K., & Magidson, J. (2021). *Upgrade manual for latent gold basic, advanced/syntax, and choice version 6.0*. Arlington, MA, Statistical Innovations Inc.

Wang, C., Xu, G., & Zhang, X. (2019). Correction for item response theory latent trait measurement error in linear mixed effects models. *Psychometrika*, *84*(3), pp673–700. https://doi.org/10.1007/s11336-019-09672-7

Wright, S. (1934). The method of path coefficients. *Annals of Mathematical Statistics*, *5*, pp161–215. https://doi.org/10.1214/aoms/1177732676

Xue, Q.-L., & Bandeen-Roche, K. (2002). Combining complete multivariate outcomes with incomplete covariate information: A latent class approach. *Biometrics*, *58*(1), pp110–120. https://doi.org/10.1111/j.0006-341X.2002.00110.x

Appendix

Table 13: Root Mean Square Error (RMSE) of the structural parameter β_3 of two one-step methods and three two-step methods in 27 settings.

		1-step		2-step	
R_m^2	R_s^2	Joint ML	Joint Bayes	ML + ML	Bayes + ML
$n = 200$					
0.5	0.0	1.56×10^{-15}	3.51×10^{-3}	1.56×10^{-15}	1.89×10^{-9}
	0.4	1.99×10^{-8}	8.798×10^{-1}	2.72×10^{-8}	1.4463
	0.8	6.32×10^{-9}	1.4352	7.49×10^{-7}	9.22×10^{-2}
0.7	0.0	5.81×10^{-16}	2.29×10^{-3}	5.81×10^{-16}	5.75×10^{-16}
	0.4	1.17×10^{-8}	7.25×10^{-3}	5.30×10^{-8}	6.551×10^{-1}
	0.8	1.57×10^{-8}	1.175×10^{-2}	5.92×10^{-7}	4.80×10^{-2}
0.9	0.0	1.51×10^{-15}	5.74×10^{-4}	1.51×10^{-15}	1.18×10^{-15}
	0.4	9.15×10^{-9}	1.67×10^{-3}	1.51×10^{-7}	7.64×10^{-4}
	0.8	1.28×10^{-8}	2.48×10^{-3}	3.56×10^{-7}	4.63×10^{-2}
$n = 600$					
0.5	0.0	6.52×10^{-16}	3.42×10^{-4}	6.52×10^{-16}	3.35×10^{-10}
	0.4	2.19×10^{-8}	6.65×10^{-3}	2.83×10^{-8}	2.86×10^{-3}
	0.8	6.64×10^{-9}	1.069×10^{-2}	6.91×10^{-7}	5.38×10^{-3}
0.7	0.0	1.72×10^{-15}	2.51×10^{-4}	1.72×10^{-15}	2.21×10^{-15}
	0.4	1.17×10^{-8}	2.27×10^{-3}	1.00×10^{-7}	1.01×10^{-3}
	0.8	1.59×10^{-8}	3.61×10^{-3}	6.04×10^{-7}	1.87×10^{-3}
0.9	0.0	1.41×10^{-15}	2.50×10^{-4}	1.41×10^{-15}	1.14×10^{-15}
	0.4	9.19×10^{-9}	5.10×10^{-4}	1.51×10^{-7}	2.93×10^{-4}
	0.8	1.18×10^{-8}	8.49×10^{-4}	3.34×10^{-7}	4.88×10^{-4}
$n = 1000$					
0.5	0.0	1.77×10^{-15}	2.49×10^{-4}	1.77×10^{-15}	1.22×10^{-15}
	0.4	2.09×10^{-8}	3.97×10^{-3}	2.84×10^{-8}	1.63×10^{-3}
	0.8	6.42×10^{-9}	6.39×10^{-3}	6.58×10^{-7}	3.16×10^{-3}
0.7	0.0	1.82×10^{-15}	1.99×10^{-4}	1.82×10^{-15}	6.42×10^{-16}
	0.4	1.23×10^{-8}	1.34×10^{-3}	6.16×10^{-8}	6.34×10^{-4}
	0.8	1.61×10^{-8}	2.20×10^{-3}	7.10×10^{-7}	1.13×10^{-3}
0.9	0.0	6.33×10^{-16}	1.65×10^{-4}	6.33×10^{-16}	5.01×10^{-16}
	0.4	9.24×10^{-9}	3.63×10^{-4}	1.47×10^{-7}	2.23×10^{-4}
	0.8	1.34×10^{-8}	5.05×10^{-4}	3.37×10^{-7}	3.53×10^{-4}

Figure 8: Estimated mean values and 95% confidence intervals of the covariance between benefDSC and costsDSC of 31 countries.

